

Bayesian Statistics

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Chapter 1

Datasets Overview

1.1 Motivating Examples

Throughout this book, we will closely examine several real-world datasets drawn from diverse fields and applications. These datasets serve as motivating examples that illustrate key concepts and methodologies discussed in later chapters. By introducing these datasets upfront, we aim to provide context and demonstrate the practical relevance of the analytical techniques you will learn. The following sections offer an overview of each dataset, highlighting their unique features and the types of insights they can yield.

1.2 Australian Institute of Sport (AIS) Dataset

The AIS Dataset [6] is a collection of biometric and physiological measurements from elite athletes at the Australian Institute of Sport. It is designed to support statistical modeling, performance analytics, and sports science research. The dataset was introduced in Cook and Weisberg [28] and remains a valuable resource for data analysis in sports and health sciences. The dataset includes diverse variables that capture blood profiles, body composition, and physical attributes of athletes, making it well-suited for regression modeling, classification, and multivariate analysis.

The dataset contains individual-level records for 202 athletes (102 male and 100 female), with each row representing a unique participant. Variables include biological sex, sporting discipline, and a range of numeric health indicators such as blood cell counts, anthropometric measurements, and fitness-related biomarkers.

The dataset contains both categorical and continuous variables, offering a rich context for exploring relationships between physiological traits and sports categories. It is widely used in teaching and research on regression modeling and multivariate graphics.

It includes:

- *Demographics*: **sex** — categorical variable (levels: female, male)

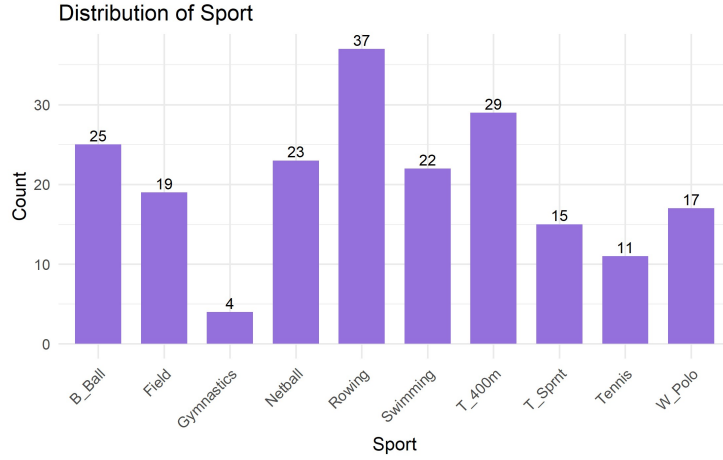


Figure 1.1: Distribution of athletes across different sports in the Australian Institute of Sport (AIS) dataset, showing the number of participants in each sport.

- *Sport Category:* **sport** — categorical variable with levels: B_Ball, Field, Gym, Netball, Row, Swim, T_400m, Tennis, T_Sprnt, W_Polo
- *Blood Measures:* Red Cell Count (RCC), White Cell Count (WCC), Hematocrit (Hc), Hemoglobin (Hg), Plasma Ferritin (Fe)
- *Body Composition:* Body Mass Index (BMI), Sum of Skin Folds (SSF), Body Fat Percentage (Bfat), Lean Body Mass (LBM)
- *Anthropometrics:* Height (Ht, cm), Weight (Wt, kg)

Figures 1.1 and 1.2 present, respectively, the distribution of athletes across different sports in the dataset and the distribution of selected numeric variables. Figure 1.1 displays a bar plot showing the number of participants in each sport, while Figure 1.2 depicts histograms of observed values with overlaid kernel density curves to illustrate the underlying distributions. Variables in Figure 1.2 are plotted on free scales to reflect their individual ranges.

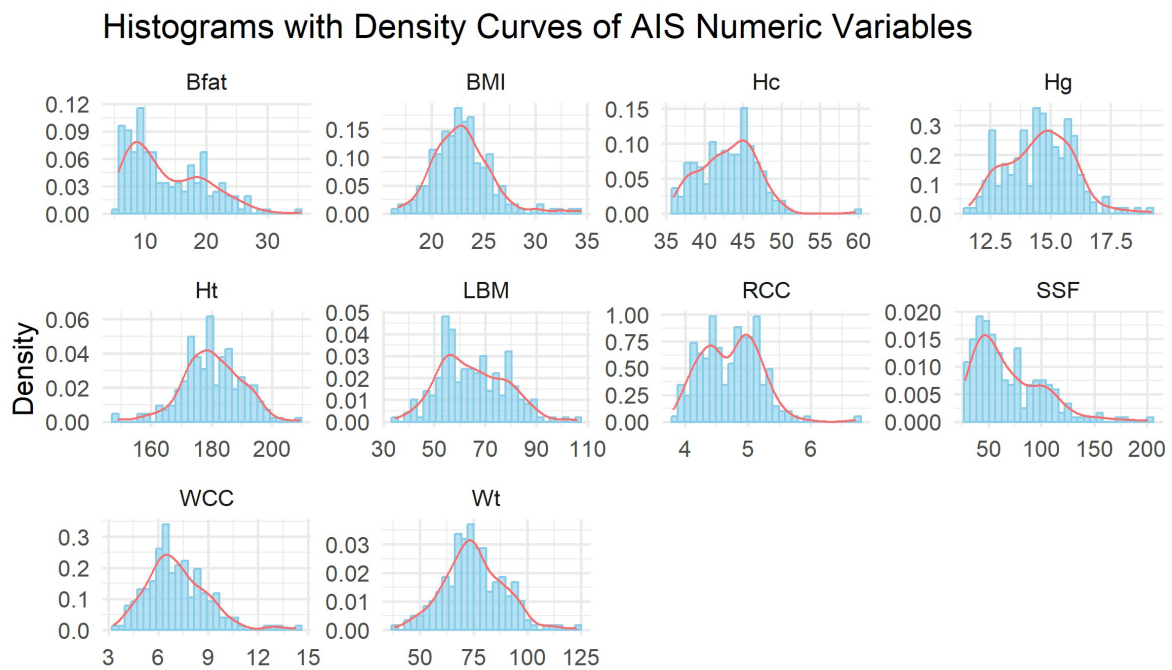


Figure 1.2: Distribution of selected numeric variables from the Australian Institute of Sport (AIS) dataset. Each panel shows a histogram of observed values with an overlaid kernel density curve to illustrate the underlying distribution.

1.3 Student Depression Dataset

The Student Depression Dataset is a multidimensional collection of survey data aimed at understanding mental health trends among students. It integrates a range of demographic, academic, and lifestyle factors to facilitate research in psychology, education, and data science. The dataset supports predictive modeling of depression and informs early intervention strategies targeting at-risk student populations.

The dataset contains individual-level records, with each row representing a unique student. Variables include demographic characteristics (e.g., age, gender, city), academic indicators (e.g., CGPA, academic pressure, study satisfaction), lifestyle and wellbeing metrics (e.g., sleep duration, dietary habits, work pressure, daily hours of study or work), and other contextual factors such as financial stress, job satisfaction, and family mental health history. The primary outcome variable is Depression Status, a binary indicator denoting whether a student is currently experiencing depression. This variable is suitable for classification modeling or binary outcome analysis.

This dataset is publicly available under the Apache License 2.0 and can be accessed on Kaggle ¹ and includes the following characteristics and variables:

- *Demographics:* Age, Gender, City
- *Academic Factors:* CGPA, Academic Pressure, Study Satisfaction
- *Lifestyle & Wellbeing:* Sleep Duration, Dietary Habits, Work Pressure, Job Satisfaction, Daily Work/Study Hours
- *Other Variables:* Profession, Degree Program, Financial Stress, Family Mental Health History, Suicidal Thoughts
- *Target Variable:* **Depression_Status** — binary indicator (Yes/No or 1/0) for whether the student is experiencing depression

Figure 1.3 presents barplots illustrating the distributions of suicidal thoughts, financial stress levels, and depression status within the dataset.

¹<https://www.kaggle.com/datasets/adilshamim8/student-depression-dataset/data>

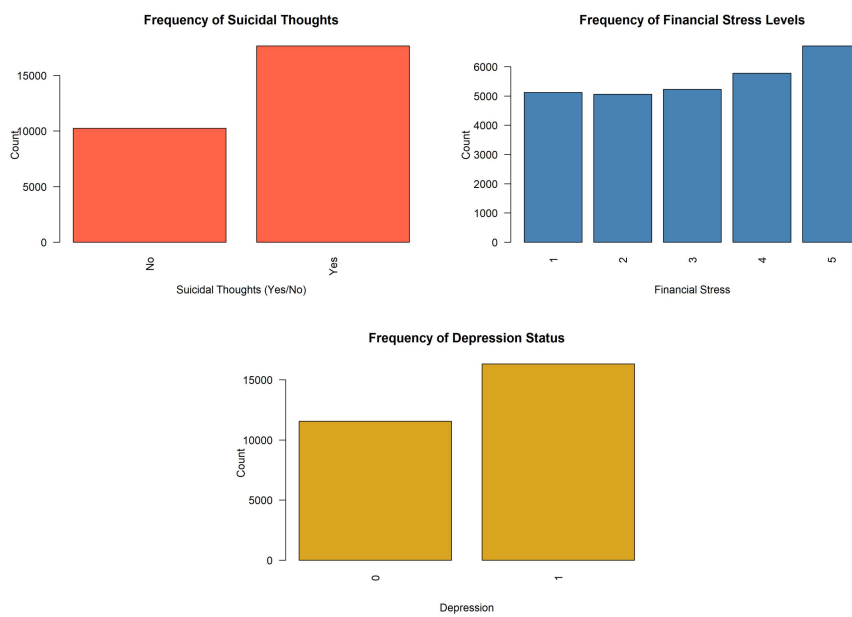


Figure 1.3: Distribution of selected variables from the Student Depression dataset. Each panel presents a barplot showing the frequency of observed values for the respective variable.

1.4 Titanic Dataset

The Titanic Dataset is a historical passenger data collection aimed at understanding survival factors from the Royal Mail Ship Titanic disaster. It integrates a variety of demographic, travel, and socio-economic variables to support research in data science, predictive modeling, and risk assessment. The dataset is widely used for binary classification tasks, such as predicting survival outcomes based on passenger characteristics. The dataset contains individual-level records, with each row representing a unique passenger. Variables include demographic attributes (e.g., age, sex), travel information (e.g., ticket class, fare, port of embarkation), and family relations aboard the ship (e.g., number of siblings/spouses, number of parents/children). The primary outcome variable is *Survived*, a binary indicator denoting whether a passenger survived the sinking. This variable is suitable for classification modeling or binary outcome analysis. This dataset is publicly available through Kaggle and can be accessed ² and includes the following characteristics and variables:

- *Demographics*: Age, Sex
- *Travel Information*: Ticket Class (Pclass), Fare, Port of Embarkation (Embarked)
- *Family Relations*: Number of siblings/spouses aboard (SibSp), Number of parents/children aboard (Parch)
- *Other Variables*: Ticket Number, Cabin Number
- *Target Variable*: **Survived** — binary indicator (1 = survived, 0 = did not survive)

Figure 1.4 illustrates the distributions of the principal numerical and categorical variables in the Titanic dataset.

²<https://www.kaggle.com/competitions/titanic/data>

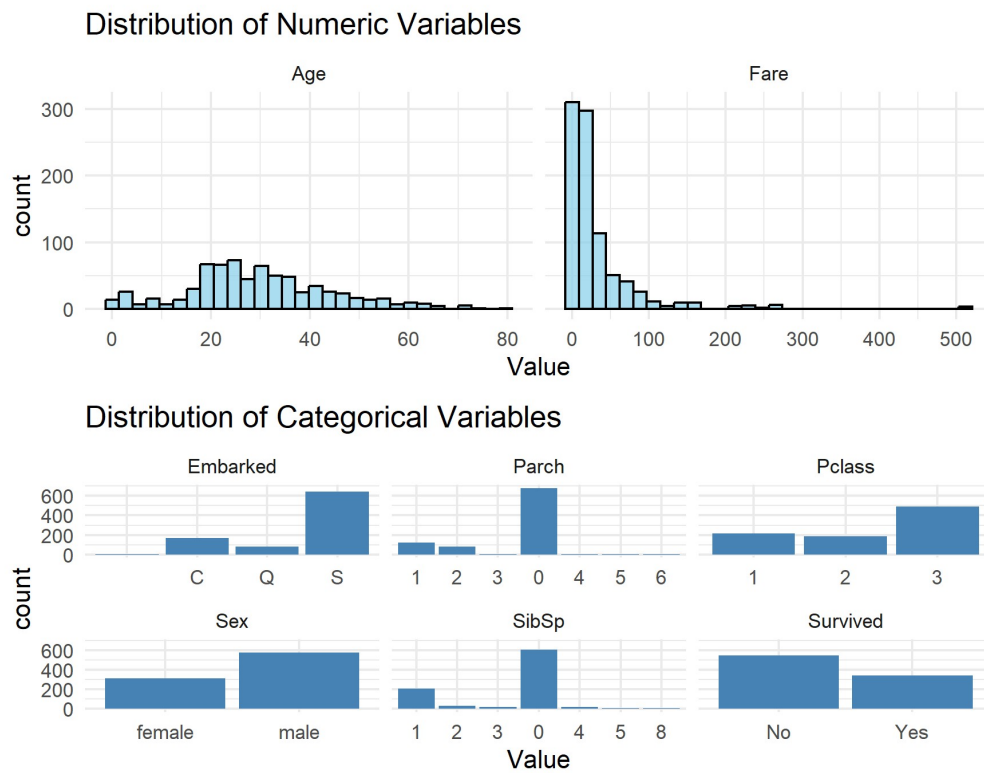


Figure 1.4: Distributions of key numeric and categorical variables from the Titanic dataset.

1.5 HIV/AIDS Dataset

The HIV/AIDS Dataset from the `jm` package [87] is a longitudinal study dataset designed to analyze the progression of HIV infection and its impact on patient survival. It supports joint modeling of biomarker trajectories and time-to-event data, commonly used in medical statistics and survival analysis. The dataset was introduced by [46] and has since been a key resource for demonstrating advanced statistical methodologies involving longitudinal and survival data³.

This dataset contains repeated measurements of CD4 cell counts, which serve as a biomarker of immune system health, taken over time for HIV-positive patients. Time points correspond to weeks since study enrollment (0, 2, 6, 12, and 18 weeks). Alongside the biomarker data, the dataset records survival times until death or censoring. The joint consideration of these two types of data allows researchers to better understand the relationship between disease progression and mortality risk.

The dataset consists of individual-level longitudinal records for multiple patients, with each row representing a single observation of a patient's CD4 count at a specific time point. Key variables include demographic, biomarker, and survival information, facilitating comprehensive modeling of disease dynamics.

It includes:

- *Patient Identifier:* `patient` — unique identifier for each patient
- *Time Variables:* `obstime` — observational time in weeks since baseline measurement
- *Biomarker:* `CD4` — CD4 cell count measurements (longitudinal)
- *Survival Outcome:* `event` — event indicator (1 = death, 0 = censored)
- *Treatment Group:* `drug` — indicator of drug treatment group
- *Other Covariates:* `gender`, `prevOI` (previous opportunistic infection), and `AZT` (Zidovudine treatment indicator)

Figures 1.5 and 1.6 present, respectively, the longitudinal trajectories of CD4 counts across patients and the Kaplan-Meier survival curve illustrating overall patient survival probabilities over time stratified by drug treatment. Figure 1.5 displays individual patient CD4 count trajectories with separate panels for each drug group, highlighting differences in immune response over the observational period. Figure 1.6 shows the Kaplan-Meier estimates of survival, allowing comparison of survival probabilities between the two drug groups.

³<https://rdrr.io/cran/joineR/man/aids.html>

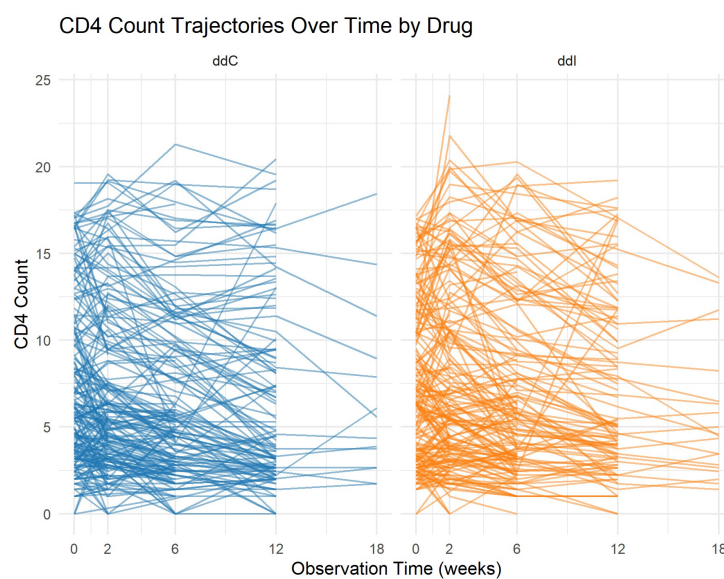


Figure 1.5: Longitudinal CD4 count trajectories for AIDS patients stratified by drug treatment. Each line represents an individual patient's CD4 count over time, with panels separated by drug group.

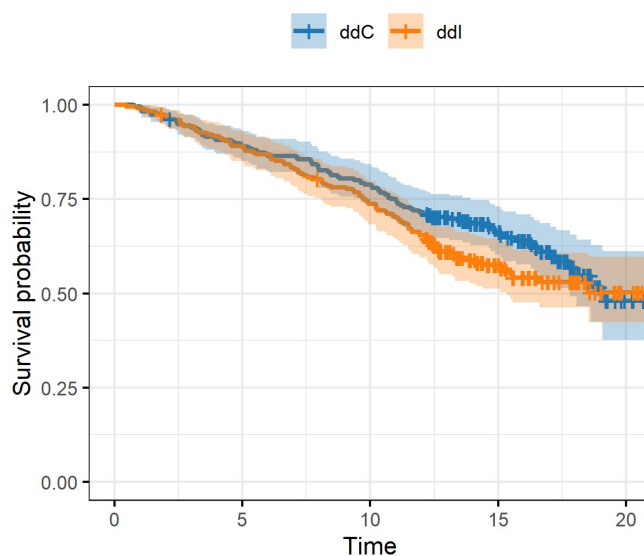


Figure 1.6: Kaplan-Meier survival curve for time to death, along with its 95% confidence bands, for HIV data.

1.6 Mayo Clinic Primary Biliary Cholangitis (PBC) Dataset

The Mayo Clinic Primary Biliary Cirrhosis (PBC) dataset [118] is a longitudinal clinical dataset collected from a randomized clinical trial conducted at the Mayo Clinic in Rochester, Minnesota, USA, beginning in 1974 and continuing into the early 1980s. The study investigated the progression of primary Biliary cirrhosis, a rare autoimmune liver disease, and evaluated the effects of D-penicillamine compared to placebo treatment.

The dataset contains 1,945 observations from 312 patients, with multiple follow-up visits per patient. Each row represents a patient visit and captures time-varying clinical and laboratory measurements. The primary outcome includes patient status, categorized as alive, transplanted, or dead, alongside a binary indicator (**status2**) for death events to support survival modeling.

Key variables include demographic factors (age, sex), clinical signs (ascites, edema, hepatomegaly), laboratory values (serum bilirubin, albumin, cholesterol, SGOT, platelets), treatment assignment, and histologic disease stage. Time is recorded via two variables: **years** indicating time to event or censoring from enrollment, and **year** representing time since enrollment at each visit.

A companion dataset, **pbc2.id**, includes only baseline measurements from the first recorded visit for each patient, and is typically used for modeling baseline survival covariates.

This dataset is widely used in statistical research and education for demonstrating Cox proportional hazards models, mixed-effects models, and joint longitudinal-survival modeling techniques. It is available in the JM package in R⁴.

It includes:

- *Demographics*: **age** (years), **sex** (male/female)
- *Clinical Signs*: **ascites**, **edema**, **hepatomegaly** — binary indicators
- *Laboratory Values*: Serum bilirubin (**bili**), albumin (**albumin**), cholesterol (**chol**), SGOT (**sgot**), platelet count (**platelet**)
- *Treatment*: **treatment** — D-penicillamine or placebo
- *Histologic Stage*: **stage** — ordinal indicator of disease severity
- *Time Variables*: **years** — time to event or censoring; **year** — time since enrollment at each visit
- *Outcome Variables*: **status** — categorical status (alive, transplanted, dead); **status2** — binary death indicator for survival analysis

Figures 1.7 present the longitudinal trajectories of serum bilirubin, albumin, serum cholesterol, SGOT, and platelets across patients, visualizing individual patient trajectories and

⁴<https://rdrr.io/cran/JM/man/pbc.html>

highlighting disease progression over time. Figure 1.8 compares cumulative incidence function estimates, which represent the probability of experiencing a specific event over time in the presence of competing risks, to assess patient survival outcomes.

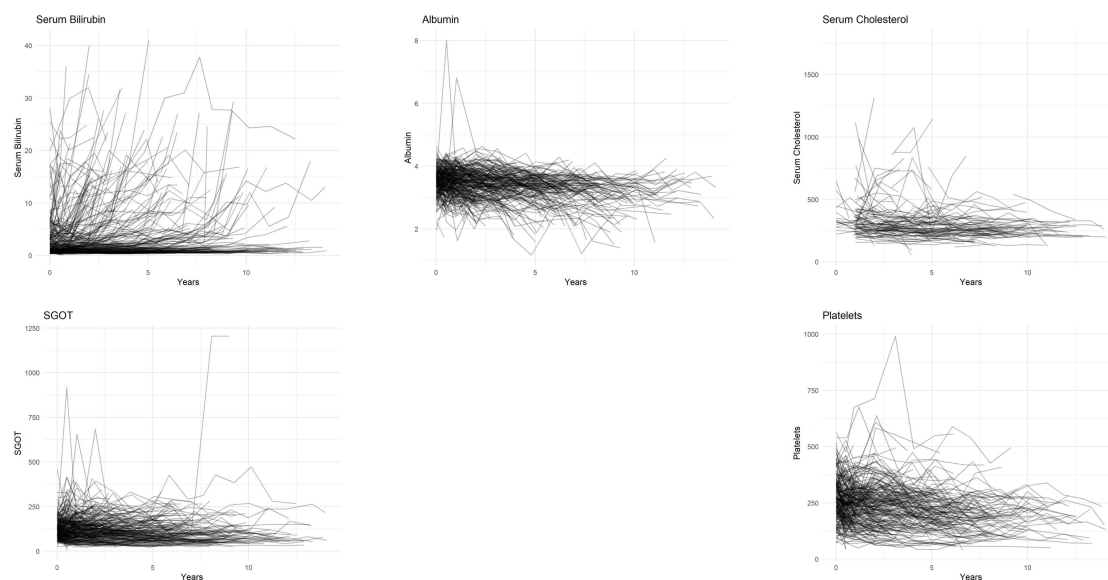


Figure 1.7: Longitudinal trajectories of key biomarkers from the PBC dataset. Individual patient trajectories over time are shown for serum bilirubin, albumin, serum cholesterol, SGOT, and platelets, illustrating the variability in disease progression among patients.

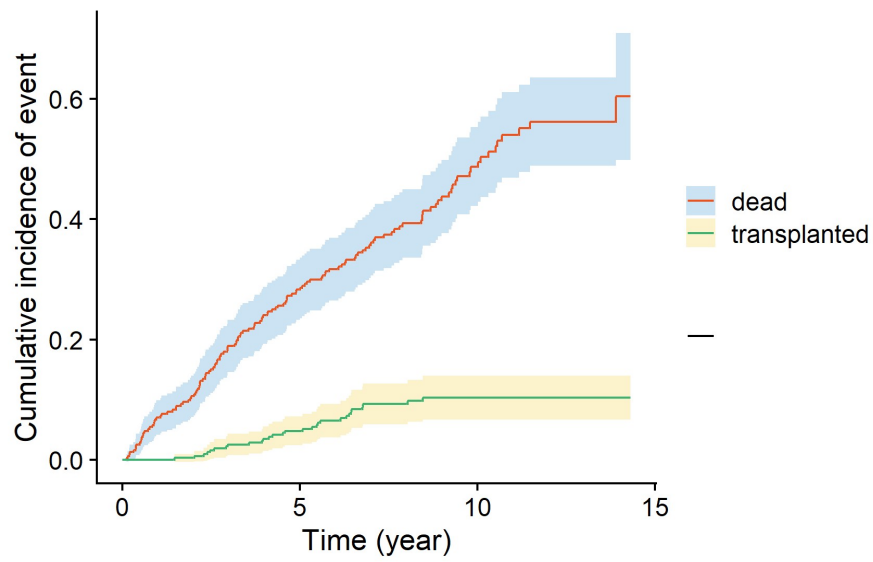


Figure 1.8: Cumulative incidence functions for competing risks of death and liver transplant among Primary Biliary Cirrhosis patients from the PBC dataset.

Chapter 2

Bayesian Inference Concepts

2.1 Introduction

In this chapter, we develop the foundational concepts of Bayesian statistical inference, beginning with the specification of prior and posterior distributions. We first introduce the role of the prior distribution in representing existing knowledge, the likelihood function in incorporating observed data, and the posterior distribution as the updated belief about unknown parameters. Both single-parameter and multiple-parameter cases are considered, with particular attention to situations where analytical simplifications arise through conjugacy. We then present methods for obtaining Bayesian estimates, including the most probable value and other estimators, along with the construction of credible intervals as Bayesian counterparts to confidence intervals. The chapter also addresses the concept of sufficiency in the Bayesian framework and concludes with an introduction to Bayesian decision theory, where inference is integrated with decision-making under uncertainty.

2.2 Prior and Posterior Distributions

Let $\mathbf{X} = (X_1, X_2, \dots, X_n)$ be a sample of n observations with joint probability mass/density function (pmf/pdf) $f_{\mathbf{X}}(\mathbf{x}|\boldsymbol{\theta})$, where $\mathbf{x} \in \mathcal{X}$ and the distribution depends on an unknown parameter $\boldsymbol{\theta}$.

2.2.1 Prior Distribution

To incorporate prior beliefs about the parameter $\boldsymbol{\theta}$ before observing the data, we assume a prior distribution $\pi(\boldsymbol{\theta})$, which represents the prior knowledge or assumptions about $\boldsymbol{\theta}$ before any data is observed. The prior can be informative, reflecting specific knowledge about the parameter, or non-informative, expressing vague or minimal prior information.

2.2.2 Likelihood Function

The likelihood function is given by

$$L(\boldsymbol{\theta}|\mathbf{x}) = f_{\mathbf{X}}(\mathbf{x}|\boldsymbol{\theta}) = \prod_{i=1}^n f_{\mathbf{X}}(x_i|\boldsymbol{\theta}),$$

which represents the probability of the observed data $\mathbf{x}$ given the parameter $\boldsymbol{\theta}$. It is important to note that the likelihood is treated as a function of $\boldsymbol{\theta}$ rather than $\mathbf{x}$. In other words, after observing the data $\mathbf{x}$, the likelihood becomes a function of the unknown parameter $\boldsymbol{\theta}$, reflecting how plausible different values of $\boldsymbol{\theta}$ are in light of the observed data.

2.2.3 Posterior Distribution

After observing the data, the prior is updated using Bayes' theorem to obtain the posterior distribution:

$$\pi(\boldsymbol{\theta}|\mathbf{x}) = \frac{f_{\mathbf{X}}(\mathbf{x}|\boldsymbol{\theta})\pi(\boldsymbol{\theta})}{\int_{\Theta} f_{\mathbf{X}}(\mathbf{x}|\boldsymbol{\theta})\pi(\boldsymbol{\theta})d\boldsymbol{\theta}}.$$

The posterior distribution reflects the updated beliefs about $\boldsymbol{\theta}$ after accounting for the observed data. The integration in the denominator may be multidimensional, as Θ denotes the parameter space.

2.2.4 Single Parameter Case

In the case where $\boldsymbol{\theta}$ reduces to a single parameter θ , the prior, likelihood, and posterior take the forms:

- Prior: $\pi(\theta)$
- Likelihood: $L(\theta|\mathbf{x}) = f_{\mathbf{X}}(\mathbf{x}|\theta)$
- Posterior: $\pi(\theta|\mathbf{x}) = \frac{f_{\mathbf{X}}(\mathbf{x}|\theta)\pi(\theta)}{m(\mathbf{x})}$, where $m(\mathbf{x}) = \int_{\Theta} f_{\mathbf{X}}(\mathbf{x}|\theta)\pi(\theta)d\theta$.

Example 1. Consider a Bernoulli distribution with success probability θ , where the prior is a Beta distribution:

$$X_i \sim \text{Bernoulli}(\theta), \quad i = 1, \dots, n, \quad \theta \sim \text{Beta}(a, b).$$

The likelihood function for n independent Bernoulli observations is:

$$f(\mathbf{x}|\theta) = \prod_{i=1}^n \theta^{x_i} (1 - \theta)^{1-x_i} = \theta^{\sum_{i=1}^n x_i} (1 - \theta)^{n - \sum_{i=1}^n x_i}.$$

Define $s = \sum_{i=1}^n x_i$ (number of successes), thus, the likelihood becomes

$$f(\mathbf{x}|\theta) = \theta^s(1 - \theta)^{n-s}.$$

The prior distribution for θ is a Beta distribution:

$$\theta \sim \text{Beta}(a, b), \quad \text{with pdf } \pi(\theta) = \frac{\theta^{a-1}(1 - \theta)^{b-1}}{B(a, b)},$$

where $B(a, b) = \int_0^1 \theta^{a-1}(1 - \theta)^{b-1} d\theta$ is the Beta function.

Using Bayes' Theorem:

$$\pi(\theta|\mathbf{x}) = \frac{f(\mathbf{x}|\theta)\pi(\theta)}{m(\mathbf{x})}.$$

Substitute the likelihood and the prior:

$$\pi(\theta|\mathbf{x}) \propto \theta^s(1 - \theta)^{n-s} \times \theta^{a-1}(1 - \theta)^{b-1}.$$

Combine the exponents:

$$\pi(\theta|\mathbf{x}) \propto \theta^{s+a-1}(1 - \theta)^{n-s+b-1}.$$

This is the kernel of a Beta distribution. Therefore, the posterior distribution is

$$\theta|\mathbf{x} \sim \text{Beta}(a + s, b + n - s).$$

Example 2. Let $X_1, X_2, \dots, X_n$ be a random sample from the normal distribution $N(\theta, \sigma^2)$, where σ^2 is known. Assume that the prior distribution for θ is normal, specifically $\theta \sim N(\mu_0, \sigma_0^2)$, that is,

$$X_i \sim N(\theta, \sigma^2), \quad i = 1, \dots, n, \quad \theta \sim N(\mu_0, \sigma_0^2).$$

Given a sample $\mathbf{x} = (x_1, \dots, x_n)$ with sample mean $\bar{x}$, the likelihood is

$$\pi(\mathbf{x}|\theta) \propto \exp\left(-\frac{n}{2\sigma^2}(\bar{x} - \theta)^2\right).$$

The prior is

$$\pi(\theta) \propto \exp\left(-\frac{(\theta - \mu_0)^2}{2\sigma_0^2}\right).$$

The posterior is proportional to the product of likelihood and prior:

$$\pi(\theta | \mathbf{x}) \propto \exp\left(-\frac{n}{2\sigma^2}(\bar{x} - \theta)^2\right) \exp\left(-\frac{1}{2\sigma_0^2}(\theta - \mu_0)^2\right).$$

Combining terms and completing the square yields

$$\pi(\theta | \mathbf{x}) \propto \exp\left(-\frac{1}{2\sigma_n^2}(\theta - \mu_n)^2\right),$$

where

$$\mu_n = \frac{\sigma_0^2 n \bar{x} + \sigma^2 \mu_0}{n \sigma_0^2 + \sigma^2}, \quad \sigma_n^2 = \frac{\sigma_0^2 \sigma^2}{n \sigma_0^2 + \sigma^2}.$$

Thus, the posterior distribution is normal:

$$\theta \mid \mathbf{x} \sim N(\mu_n, \sigma_n^2).$$

Remarks 1. The posterior mean μ_n can be interpreted as a precision-weighted average of the prior mean μ_0 and the sample mean $\bar{x}$. Specifically, the prior precision is $1/\sigma_0^2$ and the data precision is n/σ^2 , so the posterior mean balances prior information with the information from the data. Several limiting cases illustrate this relationship:

- **Noninformative prior:** If $\sigma_0^2 \rightarrow \infty$, the prior contributes negligible information, and the posterior mean reduces to the sample mean, $\mu_n \rightarrow \bar{x}$, with posterior variance $\sigma_n^2 \rightarrow \sigma^2/n$. This recovers the classical frequentist estimator.
- **Highly informative prior:** If $\sigma_0^2 \rightarrow 0$, the prior dominates the posterior, yielding $\mu_n \rightarrow \mu_0$ and $\sigma_n^2 \rightarrow 0$.
- **Large sample size:** As $n \rightarrow \infty$, the data overwhelm the prior, so $\mu_n \rightarrow \bar{x}$ and $\sigma_n^2 \sim \sigma^2/n$, regardless of the prior variance.

Exercise 1. Let $X_1, X_2, \dots, X_n$ be a random sample from the Poisson distribution with rate parameter θ and a Gamma prior:

$$X_i \sim \text{Poisson}(\theta), \quad i = 1, \dots, n, \quad \theta \sim \text{Gamma}(\alpha, \beta).$$

Prove that the posterior distribution is a Gamma distribution as follows:

$$\theta \mid \mathbf{x} \sim \text{Gamma}\left(\alpha + \sum_{i=1}^n x_i, \beta + n\right).$$

2.2.5 Multiple Parameter Case

When $\boldsymbol{\theta}$ is a vector of parameters, say $\boldsymbol{\theta} = (\theta_1, \theta_2, \dots, \theta_p)$, the prior, likelihood, and posterior become:

- Prior: $\pi(\boldsymbol{\theta})$
- Likelihood: $L(\boldsymbol{\theta} \mid \mathbf{x}) = f_{\mathbf{X}}(\mathbf{x} \mid \boldsymbol{\theta})$
- Posterior: $\pi(\boldsymbol{\theta} \mid \mathbf{x}) = \frac{f_{\mathbf{X}}(\mathbf{x} \mid \boldsymbol{\theta}) \pi(\boldsymbol{\theta})}{\int_{\Theta} f_{\mathbf{X}}(\mathbf{x} \mid \boldsymbol{\theta}) \pi(\boldsymbol{\theta}) d\boldsymbol{\theta}}$

Example 3. Consider a multivariate normal likelihood with mean vector $\boldsymbol{\mu}$ and known covariance matrix $\boldsymbol{\Sigma}$:

$$\mathbf{X} \sim N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma}), \quad \boldsymbol{\mu} \sim N_p(\boldsymbol{\mu}_0, \boldsymbol{\Sigma}_0).$$

Given that $\mathbf{X} \sim N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, the likelihood function for a single observation $\mathbf{x}$ is

$$f(\mathbf{x}|\boldsymbol{\mu}) = \frac{1}{(2\pi)^{p/2}|\boldsymbol{\Sigma}|^{1/2}} \exp\left(-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{x} - \boldsymbol{\mu})\right).$$

The prior for $\boldsymbol{\mu}$ is also a multivariate normal

$$\pi(\boldsymbol{\mu}) = \frac{1}{(2\pi)^{p/2}|\boldsymbol{\Sigma}_0|^{1/2}} \exp\left(-\frac{1}{2}(\boldsymbol{\mu} - \boldsymbol{\mu}_0)^\top \boldsymbol{\Sigma}_0^{-1}(\boldsymbol{\mu} - \boldsymbol{\mu}_0)\right).$$

Using Bayes' theorem, the posterior distribution is proportional to the product of the likelihood and the prior:

$$\pi(\boldsymbol{\mu}|\mathbf{x}) \propto f(\mathbf{x}|\boldsymbol{\mu})\pi(\boldsymbol{\mu}).$$

Substituting the likelihood and prior, we get:

$$\pi(\boldsymbol{\mu}|\mathbf{x}) \propto \exp\left(-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{x} - \boldsymbol{\mu})\right) \exp\left(-\frac{1}{2}(\boldsymbol{\mu} - \boldsymbol{\mu}_0)^\top \boldsymbol{\Sigma}_0^{-1}(\boldsymbol{\mu} - \boldsymbol{\mu}_0)\right).$$

Simplifying the exponent, we combine the terms inside the exponent:

$$\exp\left(-\frac{1}{2}\left[(\mathbf{x} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{x} - \boldsymbol{\mu}) + (\boldsymbol{\mu} - \boldsymbol{\mu}_0)^\top \boldsymbol{\Sigma}_0^{-1}(\boldsymbol{\mu} - \boldsymbol{\mu}_0)\right]\right).$$

Next, we complete the square in $\boldsymbol{\mu}$. First, expand both quadratic forms:

$$\begin{aligned} (\mathbf{x} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{x} - \boldsymbol{\mu}) &= \mathbf{x}^\top \boldsymbol{\Sigma}^{-1}\mathbf{x} - 2\boldsymbol{\mu}^\top \boldsymbol{\Sigma}^{-1}\mathbf{x} + \boldsymbol{\mu}^\top \boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}, \\ (\boldsymbol{\mu} - \boldsymbol{\mu}_0)^\top \boldsymbol{\Sigma}_0^{-1}(\boldsymbol{\mu} - \boldsymbol{\mu}_0) &= \boldsymbol{\mu}^\top \boldsymbol{\Sigma}_0^{-1}\boldsymbol{\mu} - 2\boldsymbol{\mu}^\top \boldsymbol{\Sigma}_0^{-1}\boldsymbol{\mu}_0 + \boldsymbol{\mu}_0^\top \boldsymbol{\Sigma}_0^{-1}\boldsymbol{\mu}_0. \end{aligned}$$

The total exponent becomes:

$$-\frac{1}{2}\left[\mathbf{x}^\top \boldsymbol{\Sigma}^{-1}\mathbf{x} - 2\boldsymbol{\mu}^\top \boldsymbol{\Sigma}^{-1}\mathbf{x} + \boldsymbol{\mu}^\top \boldsymbol{\Sigma}^{-1}\boldsymbol{\mu} + \boldsymbol{\mu}^\top \boldsymbol{\Sigma}_0^{-1}\boldsymbol{\mu} - 2\boldsymbol{\mu}^\top \boldsymbol{\Sigma}_0^{-1}\boldsymbol{\mu}_0 + \boldsymbol{\mu}_0^\top \boldsymbol{\Sigma}_0^{-1}\boldsymbol{\mu}_0\right].$$

Now, combine the terms involving $\boldsymbol{\mu}$:

$$-\frac{1}{2}\left[\boldsymbol{\mu}^\top (\boldsymbol{\Sigma}^{-1} + \boldsymbol{\Sigma}_0^{-1})\boldsymbol{\mu} + 2\boldsymbol{\mu}^\top (\boldsymbol{\Sigma}^{-1}\mathbf{x} + \boldsymbol{\Sigma}_0^{-1}\boldsymbol{\mu}_0) + \text{constant terms}\right].$$

To complete the square, we add and subtract the term:

$$(\boldsymbol{\Sigma}^{-1} + \boldsymbol{\Sigma}_0^{-1})^{-1} (\boldsymbol{\Sigma}^{-1}\mathbf{x} + \boldsymbol{\Sigma}_0^{-1}\boldsymbol{\mu}_0).$$

Thus, the posterior distribution has the form of a multivariate normal distribution:

$$\boldsymbol{\mu}|\mathbf{x} \sim N_p\left((\boldsymbol{\Sigma}^{-1} + \boldsymbol{\Sigma}_0^{-1})^{-1}(\boldsymbol{\Sigma}^{-1}\mathbf{x} + \boldsymbol{\Sigma}_0^{-1}\boldsymbol{\mu}_0), (\boldsymbol{\Sigma}^{-1} + \boldsymbol{\Sigma}_0^{-1})^{-1}\right).$$

Example 4. Consider a random sample drawn from a multinomial distribution with parameter vector $\boldsymbol{\theta}$, where the prior distribution for $\boldsymbol{\theta}$ follows a Dirichlet distribution:

$$\mathbf{X} \sim \text{Multinomial}(n, \boldsymbol{\theta}), \quad \boldsymbol{\theta} \sim \text{Dirichlet}(\boldsymbol{\alpha}).$$

Let $\mathbf{x} = (x_1, x_2, \dots, x_k)$ be the observed counts from a multinomial distribution with parameters n (number of trials) and $\boldsymbol{\theta} = (\theta_1, \theta_2, \dots, \theta_k)$:

$$f(\mathbf{x}|\boldsymbol{\theta}) = \frac{n!}{x_1!x_2!\dots x_k!} \prod_{i=1}^k \theta_i^{x_i}.$$

The Dirichlet prior for $\boldsymbol{\theta}$ with parameters $\boldsymbol{\alpha} = (\alpha_1, \alpha_2, \dots, \alpha_k)$ is given by:

$$\pi(\boldsymbol{\theta}) = \frac{1}{B(\boldsymbol{\alpha})} \prod_{i=1}^k \theta_i^{\alpha_i-1},$$

where $B(\boldsymbol{\alpha})$ is the multivariate Beta function defined as $B(\boldsymbol{\alpha}) = \frac{\prod_{i=1}^k \Gamma(\alpha_i)}{\Gamma(\sum_{i=1}^k \alpha_i)}$.

Using Bayes' Theorem:

$$\pi(\boldsymbol{\theta}|\mathbf{x}) \propto f(\mathbf{x}|\boldsymbol{\theta})\pi(\boldsymbol{\theta}).$$

Substitute the likelihood and prior:

$$\pi(\boldsymbol{\theta}|\mathbf{x}) \propto \left(\frac{n!}{x_1!x_2!\dots x_k!} \prod_{i=1}^k \theta_i^{x_i} \right) \left(\frac{1}{B(\boldsymbol{\alpha})} \prod_{i=1}^k \theta_i^{\alpha_i-1} \right).$$

Simplify the expression (ignoring constants that do not depend on $\boldsymbol{\theta}$):

$$\pi(\boldsymbol{\theta}|\mathbf{x}) \propto \prod_{i=1}^k \theta_i^{x_i} \prod_{i=1}^k \theta_i^{\alpha_i-1} = \prod_{i=1}^k \theta_i^{\alpha_i+x_i-1}.$$

Recognize this as the kernel of a Dirichlet distribution:

$$\boldsymbol{\theta}|\mathbf{x} \sim \text{Dirichlet}(\boldsymbol{\alpha} + \mathbf{x}).$$

Exercise 2. Normal-Inverse-Wishart Model: For a multivariate normal model with unknown mean vector and covariance matrix:

$$\mathbf{X} \sim N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma}),$$

where $\mathbf{X} \in \mathbb{R}^p$ is a p -dimensional observation, $\boldsymbol{\mu}$ is the mean vector, and $\boldsymbol{\Sigma}$ is the $p \times p$ covariance matrix. The conjugate prior is specified as:

$$\boldsymbol{\mu} | \boldsymbol{\Sigma} \sim N_p(\boldsymbol{\mu}_0, \frac{1}{\kappa_0} \boldsymbol{\Sigma}), \quad \boldsymbol{\Sigma} \sim \text{Inv-Wishart}(\nu_0, S_0),$$

where $\boldsymbol{\mu}_0$ is the prior mean vector of $\boldsymbol{\mu}$, $\kappa_0 > 0$ is a prior scaling factor for the covariance of $\boldsymbol{\mu}$, $\nu_0 > p - 1$ is the degrees of freedom of the inverse-Wishart distribution, and S_0 is a $p \times p$ positive definite scale matrix for $\boldsymbol{\Sigma}$. Then, the posterior is a normal-inverse-Wishart distribution:

$$\boldsymbol{\mu}, \boldsymbol{\Sigma} | \mathbf{x} \sim NIW(\boldsymbol{\mu}_n, \kappa_n, \nu_n, S_n).$$

where the posterior parameters are computed as follows:

$$\begin{aligned} \kappa_n &= \kappa_0 + n, \\ \boldsymbol{\mu}_n &= \frac{\kappa_0 \boldsymbol{\mu}_0 + n \bar{\mathbf{x}}}{\kappa_0 + n}, \\ \nu_n &= \nu_0 + n, \\ S_n &= S_0 + \sum_{i=1}^n (\mathbf{x}_i - \bar{\mathbf{x}})(\mathbf{x}_i - \bar{\mathbf{x}})^T + \frac{\kappa_0 n}{\kappa_0 + n} (\bar{\mathbf{x}} - \boldsymbol{\mu}_0)(\bar{\mathbf{x}} - \boldsymbol{\mu}_0)^T, \end{aligned}$$

with $\bar{\mathbf{x}} = \frac{1}{n} \sum_{i=1}^n \mathbf{x}_i$ denoting the sample mean vector.

2.3 Conjugacy

Definition 1. Let $\mathcal{F}$ denote the class of probability density/mass functions (pdf/pmf) $f(x|\theta)$, indexed by θ . A class Π of prior distributions is called a conjugate family for $\mathcal{F}$ if the posterior distribution is in the class Π for all likelihoods $f \in \mathcal{F}$, all priors $\pi \in \Pi$, and all data points $x \in X$.

In mathematical terms:

$$\text{If } f(x|\theta) \in \mathcal{F}, \pi(\theta) \in \Pi, \text{ then the posterior distribution } \pi(\theta|x) \in \Pi.$$

This condition ensures that after updating the prior $\pi(\theta)$ with the likelihood $f(x|\theta)$, the posterior distribution remains within the same class Π .

The use of conjugate priors simplifies Bayesian updating by ensuring that the posterior distribution belongs to the same family as the prior, which often leads to closed-form expressions and analytically tractable solutions.

Example 5. In Example 1, we assume that the data X_i follows a Bernoulli distribution with parameter θ , i.e., $X_i \sim \text{Bernoulli}(\theta)$ for $i = 1, \dots, n$. The prior distribution for θ is assumed to be Beta-distributed, i.e., $\theta \sim \text{Beta}(a, b)$. The posterior distribution for θ given the data $\mathbf{x}$ is then $\theta | \mathbf{x} \sim \text{Beta}(a + s, b + n - s)$, where $s = \sum_{i=1}^n x_i$ is the total number of successes observed. Therefore, the Beta distribution is conjugate to the Bernoulli likelihood in this case.

Example 6. In Example 2, we assume that the data X_i follows a normal distribution with mean θ and variance σ^2 , i.e., $X_i \sim N(\theta, \sigma^2)$, for $i = 1, \dots, n$. The prior distribution for θ is assumed to be normal, i.e., $\theta \sim N(\mu_0, \sigma_0^2)$. The posterior distribution for θ given the data $\mathbf{x}$ is then

$$\theta | \mathbf{x} \sim N \left(\frac{\sigma_0^2 \sum_{i=1}^n x_i + \sigma^2 \mu_0}{n\sigma_0^2 + \sigma^2}, \frac{\sigma_0^2 \sigma^2}{n\sigma_0^2 + \sigma^2} \right).$$

Thus, the normal distribution is conjugate to the normal likelihood in this case.

Exercise 3. In Exercise 1, prove that the Gamma distribution is conjugate to the Poisson likelihood.

2.4 Determining Bayesian Estimates

2.4.1 Maximum A Posteriori (MAP) estimation

Definition 2. The maximum a posteriori (MAP) estimate of the unknown parameter θ is the value $\hat{\theta}_{MAP}$ that maximizes the posterior distribution $\pi(\theta | \mathbf{x})$. In other words, it is the parameter value with the highest posterior probability (for a discrete parameter space) or posterior density (for a continuous parameter space). Formally,

$$\hat{\theta}_{MAP} = \arg \max_{\theta \in \Theta} \pi(\theta | \mathbf{x}),$$

or equivalently,

$$\pi(\hat{\theta}_{MAP} | \mathbf{x}) = \sup_{\theta \in \Theta} \pi(\theta | \mathbf{x}).$$

Example 7. Let $X_1, X_2, \dots, X_n$ be a random sample from the Binomial $B(m, \theta)$ distribution (m known) and assume a Beta(α, β) prior distribution for θ . Find the Bayesian most probable estimate given the sample results $\mathbf{x} = (x_1, x_2, \dots, x_n)$. The likelihood function for $X_1, X_2, \dots, X_n$ is given by:

$$L(\theta | \mathbf{x}) = \prod_{i=1}^n \binom{m}{x_i} \theta^{x_i} (1 - \theta)^{m-x_i}.$$

Since the binomial coefficients do not depend on θ , we can ignore them in the posterior distribution. Therefore, the likelihood simplifies to:

$$L(\theta | \mathbf{x}) \propto \theta^{\sum_{i=1}^n x_i} (1 - \theta)^{nm - \sum_{i=1}^n x_i}.$$

The prior distribution for θ is assumed to be a Beta distribution $B(\alpha, \beta)$. By Bayes' theorem, the posterior distribution of θ given the data $\mathbf{x} = (x_1, x_2, \dots, x_n)$ is proportional to the

product of the likelihood and the prior $\pi(\theta|\mathbf{x}) \propto L(\theta|\mathbf{x})\pi(\theta)$. Substituting the expressions for the likelihood and prior:

$$\begin{aligned}\pi(\theta | \mathbf{x}) &\propto \theta^{\sum_{i=1}^n x_i} (1 - \theta)^{nm - \sum_{i=1}^n x_i} \times \theta^{\alpha-1} (1 - \theta)^{\beta-1} \\ &\propto \theta^{\sum_{i=1}^n x_i + \alpha - 1} (1 - \theta)^{nm - \sum_{i=1}^n x_i + \beta - 1}.\end{aligned}$$

Thus, the posterior distribution is a Beta distribution:

$$\theta|\mathbf{x} \sim \text{Beta}\left(\alpha + \sum_{i=1}^n x_i, \beta + nm - \sum_{i=1}^n x_i\right).$$

The MAP estimate is the value of θ that maximizes the posterior distribution. The log of the posterior is:

$$\log(\pi(\theta|\mathbf{x})) \propto \left(\sum_{i=1}^n x_i + \alpha - 1\right) \log(\theta) + \left(nm - \sum_{i=1}^n x_i + \beta - 1\right) \log(1 - \theta).$$

Taking the derivative of the log-posterior with respect to θ , we have,

$$\frac{d}{d\theta} \log(\pi(\theta|\mathbf{x})) = \frac{\sum_{i=1}^n x_i + \alpha - 1}{\theta} - \frac{nm - \sum_{i=1}^n x_i + \beta - 1}{1 - \theta}.$$

Setting the derivative equal to zero:

$$\frac{\sum_{i=1}^n x_i + \alpha - 1}{\theta} = \frac{nm - \sum_{i=1}^n x_i + \beta - 1}{1 - \theta}$$

Solving the above equation, we get the MAP estimate:

$$\hat{\theta}_{MAP} = \frac{\sum_{i=1}^n x_i + \alpha - 1}{nm + \alpha + \beta - 2}.$$

Example 8. Let $X_1, X_2, \dots, X_n$ be a random sample from the normal distribution $N(\theta, \sigma^2)$, where σ^2 is known. Assume that the prior distribution for θ is normal, specifically $\theta \sim N(\mu_0, \sigma_0^2)$, that is,

$$X_i \sim N(\theta, \sigma^2), \quad i = 1, \dots, n, \quad \theta \sim N(\mu_0, \sigma_0^2).$$

Let us find the MAP estimate θ given the sample $\mathbf{x} = (x_1, x_2, \dots, x_n)$. As mentioned in example 2, the posterior of θ is also normal as

$$\theta|\mathbf{x} \sim N\left(\frac{\sigma_0^2 \sum_{i=1}^n x_i + \sigma^2 \mu_0}{n\sigma_0^2 + \sigma^2}, \frac{\sigma_0^2 \sigma^2}{n\sigma_0^2 + \sigma^2}\right).$$

We know that the maximum of a normal density function occurs at its mean. Therefore, the MAP estimate of θ is given by:

$$\hat{\theta}_{MAP} = \frac{\sigma_0^2 \sum_{i=1}^n x_i + \sigma^2 \mu_0}{n\sigma_0^2 + \sigma^2}.$$

This expression represents a weighted average of the sample mean $\frac{1}{n} \sum_{i=1}^n x_i$ and the prior mean μ_0 , where the weights are inversely proportional to their variances σ^2 and σ_0^2 .

Example 9. Let $X_1, X_2, \dots, X_n$ be a random sample from the Uniform distribution $U(0, \theta)$, where θ is the unknown parameter and assume a Pareto prior distribution for θ with pdf

$$\pi(\theta) = \frac{\beta}{\alpha} \left(\frac{\alpha}{\theta} \right)^{\beta+1}, \quad \alpha < \theta < \infty, \quad \alpha, \beta \text{ known}.$$

Let us to find the MAP estimate of θ given the sample $\mathbf{x} = (x_1, x_2, \dots, x_n)$. The likelihood function $L(\theta|\mathbf{x})$ is simplifies to:

$$L(\theta|\mathbf{x}) = \frac{1}{\theta^n} \prod_{i=1}^n \mathbb{I}_{[0, \theta]}(x_i) = \frac{1}{\theta^n} \mathbb{I}_{[\max(x_1, x_2, \dots, x_n), \infty)}(\theta).$$

The posterior distribution is proportional to the product of the likelihood and the prior:

$$\pi(\theta|\mathbf{x}) \propto L(\theta|\mathbf{x}) \times \pi(\theta).$$

Substituting the likelihood and prior expressions:

$$\pi(\theta|\mathbf{x}) \propto \frac{1}{\theta^n} \mathbb{I}_{[\max(x_1, x_2, \dots, x_n), \infty)}(\theta) \times \frac{1}{\theta^{\beta+1}} \mathbb{I}_{(\alpha, \infty)}(\theta).$$

Simplifying:

$$\pi(\theta|\mathbf{x}) \propto \frac{1}{\theta^{n+\beta+1}} \mathbb{I}_{[\max(\max(x_1, x_2, \dots, x_n), \alpha), \infty)}(\theta).$$

Thus, the posterior distribution is:

$$\pi(\theta|\mathbf{x}) = \begin{cases} \frac{1}{\theta^{n+\beta+1}} & \text{if } \theta \geq \max(\max(x_1, x_2, \dots, x_n), \alpha) \\ 0 & \text{otherwise.} \end{cases}$$

To find the MAP estimate of θ , we maximize the posterior distribution. Since the posterior is proportional to $\theta^{-(n+\beta+1)}$, which is a decreasing function of θ , the maximum of the posterior occurs at the smallest possible value of θ , which is:

$$\hat{\theta}_{MAP} = \max(\max(x_1, x_2, \dots, x_n), \alpha).$$

Accuracy of an estimate

Definition 3. The accuracy of an estimate $\hat{\theta}$ can be assessed by measuring how much we believe we missed the target. This can be quantified using the expected squared error:

$$\text{acc}(\hat{\theta}) = \int_{\Theta} (\hat{\theta} - \theta)^2 \pi(\theta|\mathbf{x}) d\theta, \quad (2.1)$$

where $\pi(\theta|\mathbf{x})$ is the posterior distribution of θ .

Let $\mu_p = \mathbb{E}[\theta|\mathbf{x}]$ and $\sigma_p^2 = \mathbb{V}(\theta|\mathbf{x})$ denote the mean and variance of the posterior distribution, respectively. Expanding the squared term inside the integral:

$$\begin{aligned} \text{acc}(\hat{\theta}) &= \int_{\Theta} (\hat{\theta} - \mu_p + \mu_p - \theta)^2 \pi(\theta|\mathbf{x}) d\theta \\ &= \int_{\Theta} (\hat{\theta} - \mu_p)^2 \pi(\theta|\mathbf{x}) d\theta + \int_{\Theta} (\theta - \mu_p)^2 \pi(\theta|\mathbf{x}) d\theta \\ &\quad + 2 \int_{\Theta} (\hat{\theta} - \mu_p)(\mu_p - \theta) \pi(\theta|\mathbf{x}) d\theta. \end{aligned} \quad (2.2)$$

Since the expectation of $(\mu_p - \theta)$ under the posterior distribution is zero:

$$\int_{\Theta} (\mu_p - \theta) \pi(\theta|\mathbf{x}) d\theta = 0, \quad (2.3)$$

the last term in the previous expansion vanishes, leaving:

$$\text{acc}(\hat{\theta}) = (\hat{\theta} - \mu_p)^2 + \sigma_p^2. \quad (2.4)$$

Definition 4. The accuracy of an estimate $\hat{\theta}$ of the parameter θ is given by:

$$\text{acc}(\hat{\theta}) = (\hat{\theta} - \mu_p)^2 + \sigma_p^2,$$

where $\mu_p = \mathbb{E}[\theta|\mathbf{x}]$ is the mean of the posterior distribution, and $\sigma_p^2 = \mathbb{V}(\theta|\mathbf{x})$ is the variance of the posterior distribution.

This formula highlights the trade-off between bias, represented by the squared deviation of $\hat{\theta}$ from the posterior mean, and uncertainty, quantified by the posterior variance. Essentially, the accuracy measure, we derived, corresponds to the Mean Squared Error (MSE) of the estimate $\hat{\theta}$ with respect to the posterior distribution.

2.4.2 Bayesian estimators

The expression of Definition 4 shows that the accuracy is minimized when the estimate is set to the posterior mean, i.e., $\hat{\theta} = \mu_p$. For this reason, some Bayesians prefer using the posterior mean as the estimate of θ rather than the MAP estimate, as it minimizes the expected squared error under the posterior distribution.

Definition 5. *The mean of the posterior distribution, known as the posterior mean, is given by*

$$\mu_p = \mathbb{E}(\theta \mid \mathbf{x}) = \int_{\Theta} \theta \pi(\theta \mid \mathbf{x}) d\theta.$$

If Θ is discrete, the integral is replaced by a summation. The posterior mean μ_p is the estimate of θ that minimizes the accuracy measure. For this reason, it is also referred to as the Bayes estimate of θ .

Definition 6. *Suppose $\phi(\theta)$ is a function of the parameter θ , and let φ be an estimate of $\phi(\theta)$. The accuracy of φ is defined as*

$$\text{acc}(\varphi) = \int_{\Theta} (\varphi - \phi(\theta))^2 \pi(\theta \mid \mathbf{x}) d\theta,$$

where the integral is replaced by summation if Θ is discrete.

Now, define

$$\mu_p(\phi) = \mathbb{E}(\phi(\theta) \mid \mathbf{x}) = \int_{\Theta} \phi(\theta) \pi(\theta \mid \mathbf{x}) d\theta,$$

which represents the mean (expectation) of $\phi(\theta)$ with respect to the posterior distribution of θ .

Following equation (2.4), we obtain

$$\text{acc}(\varphi) = (\varphi - \mu_p(\phi))^2 + \sigma_p^2(\phi), \quad (2.5)$$

where

$$\sigma_p^2(\phi) = \mathbb{V}(\phi(\theta) \mid \mathbf{x}) = \int_{\Theta} (\phi(\theta) - \mu_p(\phi))^2 \pi(\theta \mid \mathbf{x}) d\theta,$$

is the variance of $\phi(\theta)$ with respect to the posterior distribution of θ .

Thus, the estimate of $\phi(\theta)$ that minimizes the accuracy measure is given by

$$\hat{\phi} = \mu_p(\phi).$$

This result highlights that the posterior mean of $\phi(\theta)$ is the most accurate estimate in terms of minimizing squared error under the posterior distribution.

Definition 7. *The estimate*

$$\mu_p(\phi) = \mathbb{E}(\phi(\theta) \mid \mathbf{x}) = \int_{\Theta} \phi(\theta) \pi(\theta \mid \mathbf{x}) d\theta$$

is the estimate of $\phi(\theta)$ that minimizes the accuracy measure and is known as the Bayes estimate of $\phi(\theta)$.

Example 10. In example 1, the posterior distribution is $\theta \mid \mathbf{x} \sim \text{Beta}(a + s, b + n - s)$, so the Bayesian estimator is

$$\hat{\theta}_{\text{Bayes}} = \mathbb{E}(\theta \mid \mathbf{x}) = \frac{a + s}{a + b + n}. \quad (2.6)$$

This simplification follows from the fact that the Beta distribution is a conjugate prior for the Bernoulli likelihood, which ensures that the posterior distribution is also a Beta distribution.

This posterior mean can be interpreted as a weighted mean of the prior mean and the sample mean. This interpretation is especially clear in the case of conjugacy. For the Beta-Bernoulli model, the prior distribution is $\theta \sim \text{Beta}(a, b)$, which has a prior mean:

$$\mathbb{E}[\theta] = \frac{a}{a + b}.$$

The likelihood function for s successes in n trials follows a Binomial distribution. The sample mean is:

$$\bar{x} = \frac{s}{n}.$$

The posterior mean is given by equation 2.6. This posterior mean can be rewritten as a weighted sum of the prior mean and the sample mean:

$$\mathbb{E}[\theta \mid \mathbf{x}] = \frac{a + b}{a + b + n} \times \frac{a}{a + b} + \frac{n}{a + b + n} \times \frac{s}{n}.$$

This shows that the posterior mean is a weighted average of the prior mean and the sample mean, where the prior contributes a weight of $\frac{a+b}{a+b+n}$, reflecting the strength of the prior belief. The sample contributes a weight of $\frac{n}{a+b+n}$, which increases as more data is observed.

Example 11. In example 2, the posterior distribution is

$$\theta \mid \mathbf{x} \sim N\left(\frac{\sigma_0^2 \sum_{i=1}^n x_i + \sigma^2 \mu_0}{n\sigma_0^2 + \sigma^2}, \frac{\sigma_0^2 \sigma^2}{n\sigma_0^2 + \sigma^2}\right),$$

so the Bayesian estimator is $\hat{\theta}_{\text{Bayes}} = \frac{\sigma_0^2 \sum_{i=1}^n x_i + \sigma^2 \mu_0}{n\sigma_0^2 + \sigma^2}$.

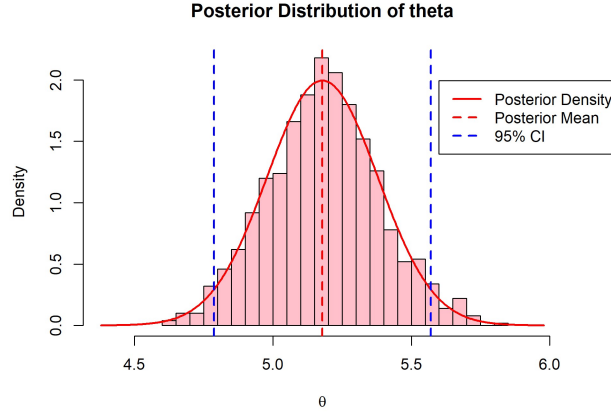


Figure 2.1: Histogram of simulated posterior data and posterior distribution of θ .

Also, this result follows from the fact that the normal distribution is a conjugate prior for the normal likelihood.

The posterior mean can be interpreted as a weighted mean of the prior mean (μ_0) and the sample mean ($\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$). Specifically:

$$\hat{\theta}_{Bayes} = \frac{\sigma_0^2 \sum_{i=1}^n x_i + \sigma^2 \mu_0}{n\sigma_0^2 + \sigma^2} = \frac{n\sigma_0^2}{n\sigma_0^2 + \sigma^2} \bar{x} + \frac{\sigma^2}{n\sigma_0^2 + \sigma^2} \mu_0.$$

The prior mean contributes a weight of $\frac{\sigma^2}{n\sigma_0^2 + \sigma^2}$, and the sample mean, $\bar{x}$, contributes a weight of $\frac{n\sigma_0^2}{n\sigma_0^2 + \sigma^2}$. Thus, the posterior mean is a weighted average of the prior mean μ_0 and the sample mean, with the weights depending on the variances σ_0^2 and σ^2 as well as the sample size n .

Remarks 2. In cases where the posterior distribution is not analytically tractable, numerical methods such as Markov Chain Monte Carlo (MCMC) or variational inference may be employed.

Numerical Example 1. In Example 2, generate a sample of size $n = 100$ from a normal distribution with $\theta = 5$ and $\sigma = 2$. Set the hyperparameters for the priors as $\mu_0 = 0$ and $\sigma_0 = 10$. Draw a histogram of the generated data, as well as a histogram of the posterior distribution. Highlight the interval containing 95% of the posterior probability.

Figure 2.1 shows the histogram of the posterior sample of θ , with the posterior density in red, the posterior mean as a red dashed line, and the 95% credible interval in blue dashed lines.

2.5 Credible intervals

In Bayesian statistics, an interval or set estimate is referred to as a credible interval or Bayesian confidence interval. It is defined as follows:

Definition 8. A $100(1 - \alpha)\%$ credible interval (or set) for the parameter θ , given data $\mathbf{x}$, is a set $C_{\mathbf{x}}$ such that

$$\int_{C_{\mathbf{x}}} \pi(\theta | \mathbf{x}) d\theta = 1 - \alpha.$$

Given the posterior distribution of θ given $\mathbf{X} = \mathbf{x}$, we would like to find the set $C_{\mathbf{x}}$ that satisfies

$$\int_{C_{\mathbf{x}}} \pi(\theta | \mathbf{x}) d\theta = 1 - \alpha, \quad (2.7)$$

$$\text{Size}(C_{\mathbf{x}}) \leq \text{Size}(C'_{\mathbf{x}}), \quad (2.8)$$

for any set $C'_{\mathbf{x}}$ satisfying $\int_{C'_{\mathbf{x}}} \pi(\theta | \mathbf{x}) d\theta \geq 1 - \alpha$. The set $C_{\mathbf{x}}$ is therefore the one with the smallest possible size (or, in one dimension, the shortest length) among all such sets.

If we take length as our measure of size, we obtain the following result.

Definition 9 (Highest Posterior Density (HPD) Credible Set). Suppose the posterior density $\pi(\theta | \mathbf{x})$ is unimodal. For a given $\alpha \in (0, 1)$, the shortest credible interval for θ is given by

$$C_{\mathbf{x}} = \{\theta : \pi(\theta | \mathbf{x}) \geq k\}, \quad (2.9)$$

where the constant k is chosen such that

$$\int_{C_{\mathbf{x}}} \pi(\theta | \mathbf{x}) d\theta = 1 - \alpha. \quad (2.10)$$

The set $C_{\mathbf{x}}$ is called a highest posterior density (HPD) set, since it consists of the parameter values with the highest posterior density.

Remarks 3. The HPD credible interval is not necessarily a single continuous interval; it may consist of multiple disjoint intervals, depending on the shape of the posterior distribution.

Remarks 4. In some cases, determining a $100(1 - \alpha)\%$ HPD credible interval can be challenging and may require a numerical search approach. A practical and often more feasible alternative is to compute a $100(1 - \alpha)\%$ equal-tailed credible interval. This interval, denoted as $C_{\mathbf{x}}$, is defined as (θ_L, θ_U) , where

$$\int_{-\infty}^{\theta_L} \pi(\theta | \mathbf{x}) d\theta = \int_{\theta_U}^{\infty} \pi(\theta | \mathbf{x}) d\theta = \frac{\alpha}{2}.$$

This ensures that an equal probability of $\frac{\alpha}{2}$ is excluded from each tail of the posterior distribution, making it a straightforward alternative to the HPD interval when direct computation is difficult.

Example 12. Let $X_1, X_2, \dots, X_n$ be a random sample from the Poisson distribution with rate parameter θ , and suppose we have a Gamma prior for θ :

$$X_i \sim \text{Poisson}(\theta), \quad i = 1, \dots, n, \quad \theta \sim \text{Gamma}(\alpha, \beta).$$

Let us find the posterior distribution of θ and derive both the equal-tailed credible interval and the highest posterior density (HPD) interval.

The posterior distribution can be expressed as:

$$\pi(\theta|\mathbf{x}) \propto \theta^{(\sum x_i + \alpha) - 1} e^{-(n + \beta)\theta}.$$

Recognizing this as the kernel of a Gamma distribution, we conclude:

$$\theta|\mathbf{x} \sim \text{Gamma}(\alpha + \sum x_i, \beta + n),$$

An equal-tailed credible interval for θ is given by the quantiles of the posterior Gamma distribution:

$$P(\theta_L \leq \theta \leq \theta_U|\mathbf{x}) = 1 - \alpha.$$

where θ_L and θ_U satisfy:

$$P(\theta \leq \theta_L|\mathbf{x}) = \frac{\alpha}{2}, \quad P(\theta \geq \theta_U|\mathbf{x}) = \frac{\alpha}{2}.$$

This means:

$$\theta_L = F^{-1}\left(\frac{\alpha}{2}; \alpha^*, \beta^*\right), \quad \theta_U = F^{-1}\left(1 - \frac{\alpha}{2}; \alpha^*, \beta^*\right),$$

where F^{-1} is the inverse cumulative distribution function (CDF) of the Gamma distribution with updated parameters $\alpha^* = \alpha + \sum_{i=1}^n x_i$ and $\beta^* = \beta + n$. The HPD credible interval is the shortest interval containing $1 - \alpha$ probability mass. Since the Gamma distribution is unimodal, the HPD interval is given by the values satisfying:

$$\int_{\theta_{HPD,L}}^{\theta_{HPD,U}} \pi(\theta|\mathbf{x}) d\theta = 1 - \alpha.$$

where $\theta_{HPD,L}$ and $\theta_{HPD,U}$ are determined such that:

$$\pi(\theta_{HPD,L}|\mathbf{x}) = \pi(\theta_{HPD,U}|\mathbf{x}).$$

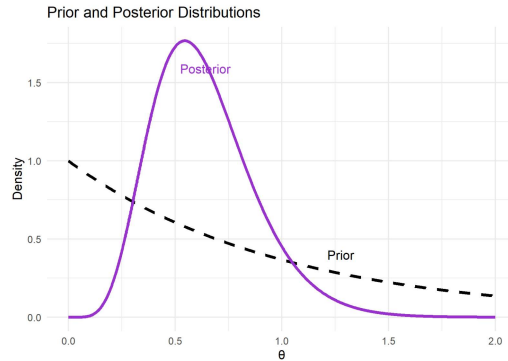


Figure 2.2: Comparison of the prior and posterior distributions for the Poisson-Gamma model with parameters $\alpha = \beta = 1$, $n = 10$, and $\sum_{i=1}^n x_i = 6$. The prior distribution is shown as a dashed line, while the posterior distribution is displayed as a solid line.

Numerical Example 2. In Example 12, let $\alpha = \beta = 1$, $n = 10$, and $\sum_{i=1}^n x_i = 6$. We compare the 95% equal-tailed credible interval with that of the HPD interval. Figure 2.2 illustrates the prior and posterior distributions for this dataset, while Figure 2.3 depicts the posterior distribution along with the HPD and equal-tailed credible intervals.¹ The equal-tailed interval is (0.256, 1.187) with width 0.931, while the HPD interval is (0.214, 1.114) with width 0.900. Thus, the HPD interval is shorter, as expected.

¹The R code used to generate this figure is available at <https://github.com/tbaghfalaki/Bayesian-Statistics/blob/main/Chapter2/NE1.R>

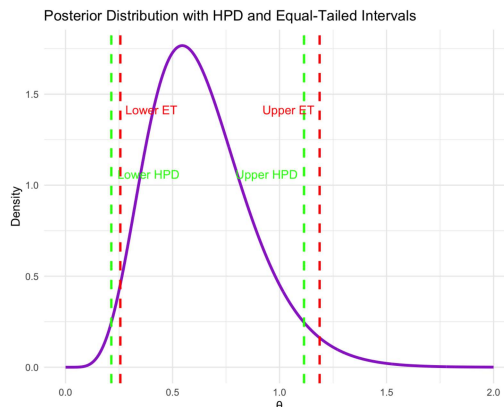


Figure 2.3: Posterior distribution of θ for the Poisson-Gamma model with parameters $\alpha = \beta = 1$, $n = 10$, and $\sum_{i=1}^n x_i = 6$ with the 95% highest posterior density (HPD) interval (green dashed lines) and the 95% equal-tailed credible interval (red dashed lines).

Numerical Example 3. Consider example 2 with $n = 10$ sample with $\bar{x} = 2.5$ and $\sigma^2 = 4$ and $\mu_0 = 0$ such that $\sigma_0^2 \in \{0.1, 1, 10, 100, 1000\}$. Figure 2.4 shows a comparison of the prior and posterior distributions for this model. This plot demonstrates that as σ_0^2 increases, the influence of the prior diminishes, and the posterior distribution becomes more concentrated around the sample mean $\bar{x}$. This reflects the increasing dominance of the likelihood over the prior in determining the posterior distribution.

Also, Figure 2.5 illustrates the HPD and equal-tailed credible intervals, where the HPD interval is computed using the *HDInterval* package in R. Based on this figure, it is evident that for $\sigma_0^2 > 100$, the credible intervals remain unchanged.

The R code used to generate this figure is available at <https://github.com/tbaghfalaki/Bayesian-Statistics/blob/main/Chapter2/NE2.R>.

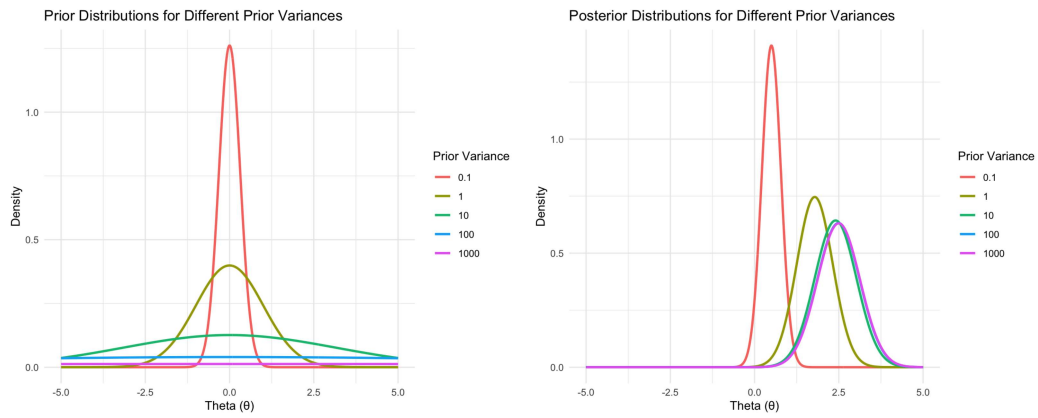


Figure 2.4: Comparison of the prior and posterior distributions for the normal-normal model with parameters $\mu_0 = 0$ and $\sigma_0^2 \in \{0.1, 1, 10, 100, 1000\}$ for $n = 10$ sample with $\bar{x} = 2.5$.

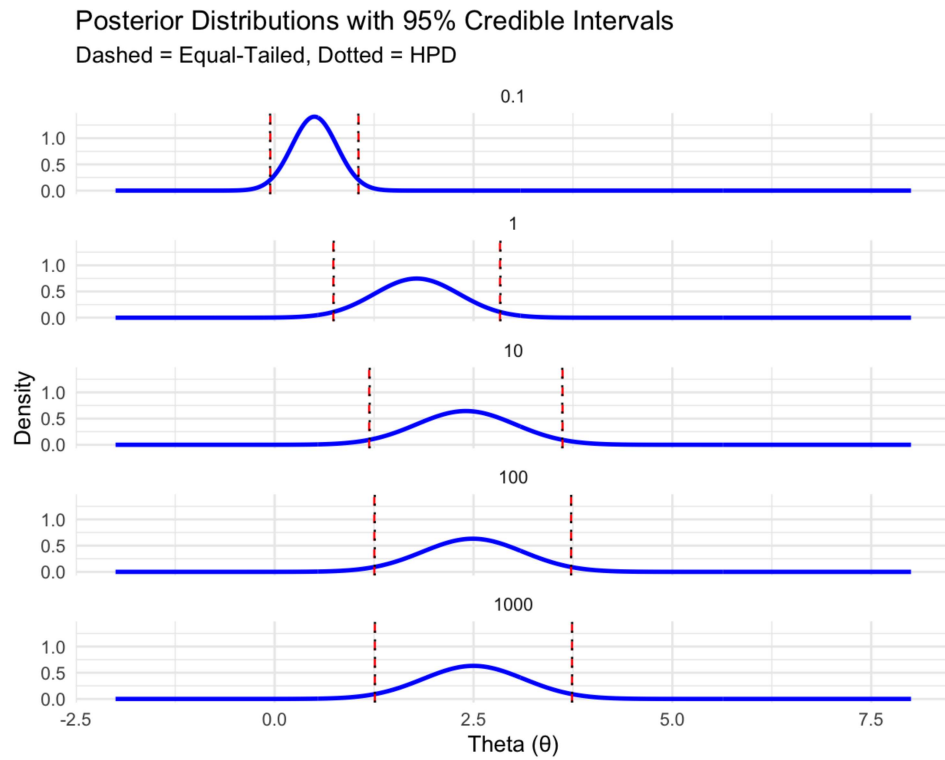


Figure 2.5: Posterior distributions of θ for different prior variances. Dashed vertical lines indicate the 95% equal-tailed credible intervals, while dotted vertical lines show the 95% highest posterior density (HPD) intervals. The figure illustrates how the posterior mean and credible intervals shrink toward the prior as the prior variance decreases.

2.6 Sufficiency in Bayesian Statistics

In this section, we investigate the effect of using sufficient statistics instead of the full observed data in Bayesian analysis. We begin with a brief review: a sufficient statistic is a function of the data that retains all information about the parameter. To formalize this, we review the Factorization Theorem for sufficient statistics.

Theorem 1 (Factorization Theorem). *A statistic $T(\mathbf{X})$ is sufficient for the parameter θ if and only if the likelihood function can be factorized as*

$$f_{\mathbf{X}}(\mathbf{x} | \theta) = g(T(\mathbf{x}), \theta) h(\mathbf{x})$$

for all $\mathbf{x} \in \mathcal{X}_{(n)}$, where $T(\mathbf{x})$ is the value of the statistic.

In other words, the likelihood function can be expressed as the product of two functions: $g(T(\mathbf{x}), \theta)$, which depends on the parameter θ and the sufficient statistic $T(\mathbf{x})$, and $h(\mathbf{x})$, which does not depend on θ .

Example 13. Consider a random sample $X_1, X_2, \dots, X_n$ from a Bernoulli distribution with parameter θ , where $X_i \sim \text{Bern}(\theta)$, the likelihood function is given by

$$f_{\mathbf{X}}(\mathbf{x}|\theta) = \theta^{\sum x_i} (1 - \theta)^{n - \sum x_i}.$$

Define the statistic $T(\mathbf{x}) = \sum_{i=1}^n X_i$, which represents the total number of successes. The likelihood function can now be rewritten as

$$f_{\mathbf{X}}(\mathbf{x}|\theta) = \left[\theta^{T(\mathbf{x})} (1 - \theta)^{n - T(\mathbf{x})} \right] \cdot 1.$$

Comparing this with the factorization theorem form $f_{\mathbf{X}}(\mathbf{x}|\theta) = g(T(\mathbf{x}), \theta)h(\mathbf{x})$, we can identify:

$$g(T(\mathbf{x}), \theta) = \theta^{T(\mathbf{x})} (1 - \theta)^{n - T(\mathbf{x})},$$

which depends on θ and the statistic $T(\mathbf{x})$, and $h(\mathbf{x}) = 1$, which is independent of θ . Since the likelihood function factors in this form, the statistic $T(\mathbf{x}) = \sum X_i$ is sufficient for θ by the Factorization Theorem.

Theorem 2. *If $T(\mathbf{X})$ is a sufficient statistic, then for $t = T(\mathbf{x})$, the posterior distribution satisfies*

$$\pi(\theta|\mathbf{x}) = \pi(\theta|t).$$

This implies that the posterior distribution of θ depends on the data $\mathbf{x}$ only through the sufficient statistic $T(\mathbf{x})$.

Proof. If $T(x)$ is sufficient for θ , then by the Factorization Theorem, the likelihood function can be written as

$$f_X(x|\theta) = g(T(x), \theta)h(x),$$

where $g(T(x), \theta)$ depends on θ and the statistic $T(x)$, while $h(x)$ is independent of θ . Using Bayes' theorem, the posterior distribution of θ given x is

$$\pi(\theta|x) = \frac{f_X(x|\theta)\pi(\theta)}{f_X(x)}.$$

Substituting the factorized form,

$$\pi(\theta|x) = \frac{g(T(x), \theta)h(x)\pi(\theta)}{f_X(x)}.$$

The marginal likelihood is obtained by integrating out θ : $f_X(x) = \int f_X(x|\theta)\pi(\theta)d\theta$. Substituting the factorized likelihood,

$$f_X(x) = \int g(T(x), \theta)h(x)\pi(\theta)d\theta.$$

Since $h(x)$ is independent of θ , we have:

$$f_X(x) = h(x) \int g(T(x), \theta)\pi(\theta)d\theta.$$

Defining

$$f_T(T(x)) = \int g(T(x), \theta)\pi(\theta)d\theta,$$

we obtain:

$$f_X(x) = h(x)f_T(T(x)).$$

Substituting this into the posterior expression,

$$\pi(\theta|x) = \frac{g(T(x), \theta)h(x)\pi(\theta)}{h(x)f_T(T(x))}.$$

Canceling $h(x)$,

$$\pi(\theta|x) = \frac{g(T(x), \theta)\pi(\theta)}{f_T(T(x))}.$$

Since

$$\pi(\theta|T(x)) = \frac{f_T(T(x)|\theta)\pi(\theta)}{f_T(T(x))},$$

and $f_T(T(x)|\theta) = g(T(x), \theta)$, it follows that

$$\pi(\theta|x) = \pi(\theta|T(x)).$$

Thus, the posterior distribution of θ depends on x only through the sufficient statistic $T(x)$, completing the proof. $\square$

Remarks 5. Since the existence of a sufficient statistic $T(\mathbf{X})$ for θ ensures that the posterior distribution of θ given the sample $\mathbf{X} = \mathbf{x}$ depends on the data only through $t = T(\mathbf{x})$, all estimation procedures discussed previously can be performed using the posterior density (or mass function)

$$\pi(\theta \mid T(\mathbf{X}) = t)$$

instead of

$$\pi(\theta \mid \mathbf{X} = \mathbf{x}).$$

For example, the MAP estimate $\hat{\theta}$ can equivalently be expressed as

$$\hat{\theta} = \arg \max_{\theta \in \Theta} \pi(\theta \mid \mathbf{X} = \mathbf{x}) = \arg \max_{\theta \in \Theta} \pi(\theta \mid T(\mathbf{X}) = t),$$

where $t = T(\mathbf{x})$ is the value of the sufficient statistic T .

Example 14. In Example 2, the sufficient statistic is $\bar{X}$. Since $X_i \sim N(\theta, \sigma^2)$, $i = 1, 2, \dots, n$, the sample mean follows

$$\bar{X} \sim N\left(\theta, \frac{\sigma^2}{n}\right).$$

Using Bayesian updating, the posterior distribution is given by

$$\pi(\theta \mid \bar{x}) \propto f(\bar{x} \mid \theta) \pi(\theta) \propto \exp\left(-\frac{1}{2} \left[\frac{n}{\sigma^2} (\theta - \bar{x})^2 + \frac{1}{\sigma_0^2} (\theta - \mu_0)^2 \right]\right).$$

From this expression, the posterior distribution follows a normal distribution:

$$\theta \mid \bar{x} \sim N(\mu_n, \sigma_n^2),$$

where the updated mean and variance are

$$\mu_n = \frac{\frac{n}{\sigma^2} \bar{x} + \frac{1}{\sigma_0^2} \mu_0}{\frac{n}{\sigma^2} + \frac{1}{\sigma_0^2}}, \quad \sigma_n^2 = \frac{1}{\frac{n}{\sigma^2} + \frac{1}{\sigma_0^2}}.$$

This result is identical to what we obtained in Example 2 when using the entire sample, demonstrating that the sufficient statistic $\bar{x}$ retains all necessary information for Bayesian estimation of θ .

2.7 Bayesian Decision Theory

Bayesian Decision Theory is a fundamental statistical framework for decision-making under uncertainty, grounded in probability theory and utility theory. It models decisions by combining prior beliefs (prior probabilities), observed data (likelihood), and potential

outcomes—quantified through either utilities (benefits) or losses (costs)—to determine the optimal action. When using a loss function, the goal is to minimize cost: the smaller the loss, the better. Conversely, when using a utility function, the objective is to maximize benefit: the higher the utility, the better. The core principle is to choose the action that either maximizes expected utility or minimizes expected loss, calculated by averaging the utilities or losses of possible outcomes, weighted by their posterior probabilities. This framework is widely applied across fields such as statistics, machine learning, economics, and artificial intelligence for tasks involving classification, prediction, estimation, and optimization under uncertainty. In statistics specifically, Bayesian Decision Theory plays a key role in parameter estimation, hypothesis testing, and model selection, where decisions must balance the evidence from data with prior beliefs and the consequences of different choices.

Definition 10. A loss function is a mathematical function $L(a, \theta)$

$$L : \mathcal{A} \times \Theta \rightarrow \mathbb{R}_{\geq 0},$$

that assigns a numerical cost to each pair (a, θ) , where $a \in \mathcal{A}$ is an action from the action space and $\theta \in \Theta$ is the true state of nature. It quantifies the penalty incurred by choosing action a when the actual state is θ , providing a basis for evaluating and comparing decision-making strategies under uncertainty.

Types of Loss Functions

1. Zero-One Loss (0-1 Loss): The 0-1 loss function is used primarily for classification problems. It takes the value 1 if the chosen action is incorrect and 0 if it is correct:

$$L(a, \theta) = \begin{cases} 0 & \text{if } a = \theta, \\ 1 & \text{if } a \neq \theta, \end{cases}$$

where a is the chosen class label and θ is the true class.

2. Squared Error Loss (L2 Loss): Squared error loss is commonly used in regression problems. It penalizes the squared difference between the chosen action and the true parameter value:

$$L(a, \theta) = (a - \theta)^2.$$

3. Absolute Error Loss (L1 Loss): The absolute error loss measures the absolute difference between the chosen action and the true parameter:

$$L(a, \theta) = |a - \theta|.$$

4. Huber Loss: Huber loss combines squared error loss and absolute error loss. It is

quadratic for small deviations and linear for large deviations:

$$L(a, \theta) = \begin{cases} \frac{1}{2}(a - \theta)^2 & \text{if } |a - \theta| \leq \delta, \\ \delta|a - \theta| - \frac{1}{2}\delta^2 & \text{if } |a - \theta| > \delta. \end{cases}$$

5. Logarithmic Loss (Log Loss): Log loss is commonly used in binary classification problems. It penalizes actions that assign probability away from the true label:

$$L(a, \theta) = -(\theta \log(a) + (1 - \theta) \log(1 - a)),$$

where a is the action representing the predicted probability of the positive class, and $\theta \in \{0, 1\}$ is the true label.

6. Cross-Entropy Loss: Cross-entropy loss generalizes log loss to multi-class classification:

$$L(a, \theta) = -\sum_i \theta_i \log(a_i),$$

where θ_i is the true distribution (usually a one-hot vector) and a_i is the action representing the predicted probability for each class.

7. Exponential Loss: Exponential loss is used in boosting algorithms (e.g., AdaBoost). It grows exponentially with the magnitude of the prediction error:

$$L(a, \theta) = \exp(-\theta a),$$

where $\theta \in \{-1, +1\}$ is the true class label and a is the chosen action (e.g., margin output by a classifier).

8. Pinball Loss (Quantile Loss): Pinball loss is used in quantile regression. It measures the weighted difference between the chosen action and the true value:

$$L_\tau(a, \theta) = \begin{cases} \tau(\theta - a) & \text{if } a < \theta, \\ (1 - \tau)(a - \theta) & \text{if } a \geq \theta, \end{cases}$$

where $\tau \in (0, 1)$ is the target quantile (e.g., $\tau = 0.5$ for the median).

9. Misclassification Loss: Misclassification loss is used in classification problems where the action is a chosen class label:

$$L(a, \theta) = \begin{cases} 0 & \text{if } a \text{ is the correct class,} \\ 1 & \text{if } a \text{ is the incorrect class.} \end{cases}$$

Figure 2.6 displays five different loss functions, each representing how the loss varies with respect to the predicted value a when the true value $\theta = 1$. The Zero-One Loss is a binary step function, assigning a loss of 1 for incorrect predictions and 0 for correct ones. Squared Error Loss increases quadratically, penalizing larger deviations more heavily. Absolute

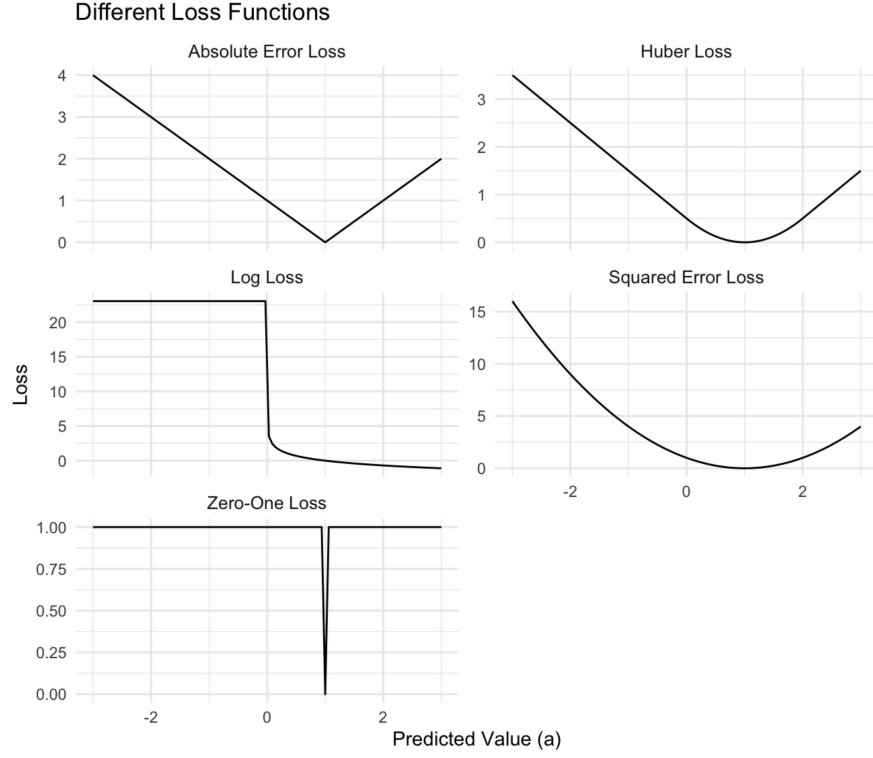


Figure 2.6: Comparison of different loss functions: Zero-One Loss, Squared Error Loss, Absolute Error Loss, Huber Loss, and Log Loss. The x-axis represents the predicted value a , and the y-axis shows the corresponding loss when the true value $\theta = 1$.

Error Loss grows linearly, providing a more robust measure against outliers. Huber Loss combines the behaviors of squared and absolute errors, being quadratic for small errors and linear for large ones, thus balancing sensitivity and robustness. Finally, Log Loss shows a logarithmic curve, penalizing predictions closer to 0 or 1 more heavily, and is commonly used in binary classification tasks. Each loss function highlights different trade-offs in terms of error sensitivity, robustness, and suitability for regression or classification problems.

Definition 11. *Risk function:* Given an action $a \in \mathcal{A}$, a parameter $\theta \in \Theta$, and a loss function $L(a, \theta)$, the risk function $R(a, \theta)$ is defined as the expected loss under the probability distribution determined by θ :

$$R(a, \theta) = \mathbb{E}_{\theta} [L(a, \theta)] = \int L(a, \theta) f(\mathbf{x} \mid \theta) d\mathbf{x},$$

Here, $\mathbb{E}_{\theta}$ denotes the expectation taken with respect to the distribution of the data when the true parameter is θ .

In practice, the best estimator (or decision rule) is the one that minimizes the risk function. That is,

$$\hat{a} = \arg \min_{a \in \mathcal{A}} R(a, \theta).$$

Example 15. [25] Let $\delta(\mathbf{X})$ be an estimator of the parameter θ , and consider the mean squared error (MSE), defined as

$$MSE(\theta) = \mathbb{E}_\theta \left[(\delta(\mathbf{X}) - \theta)^2 \right].$$

This expression is equivalent to the risk function $R(\theta, \delta)$ when the loss function is the squared error loss, $L(\theta, a) = (a - \theta)^2$; that is,

$$MSE(\theta) = \mathbb{E}_\theta L(\theta, \delta(\mathbf{X})) = R(\theta, \delta).$$

Under squared error loss, the risk function decomposes as

$$R(\theta, \delta) = \text{Var}_\theta (\delta(\mathbf{X})) + (\mathbb{E}_\theta [\delta(\mathbf{X})] - \theta)^2 = \text{Var}_\theta (\delta(\mathbf{X})) + (\text{Bias}_\theta (\delta(\mathbf{X})))^2. \quad (2.11)$$

Example 16. [25] Let $X_1, \dots, X_n$ be a random sample from a $N(\mu, \sigma^2)$ population. Consider estimating σ^2 using squared error loss. We will consider estimators of the form $\delta_b(\mathbf{X}) = bS^2$, where S^2 is the sample variance and b can be any nonnegative constant. Recall that $ES^2 = \sigma^2$ and, for a normal sample, $\text{Var} S^2 = \frac{2\sigma^4}{n-1}$. Using (2.11), we can compute the risk function for $\delta_b(\mathbf{X})$ as

$$\begin{aligned} R((\mu, \sigma^2), \delta_b(\mathbf{X})) &= \text{Var}(bS^2) + (\mathbb{E}[bS^2] - \sigma^2)^2 \\ &= b^2 \text{Var}(S^2) + (b \mathbb{E}[S^2] - \sigma^2)^2 \\ &= \frac{2b^2\sigma^4}{n-1} + (b-1)^2\sigma^4 \quad (\text{since } \text{Var}(S^2) = \frac{2\sigma^4}{n-1}) \\ &= \left[(b-1)^2 + \frac{2b^2}{n-1} \right] \sigma^4. \end{aligned}$$

To find the optimal value of b that minimizes the risk, we minimize the expression in brackets:

$$f^*(b) = \frac{2b^2}{n-1} + (b-1)^2.$$

Thus, the value of b that minimizes the risk is $b^* = \frac{n-1}{n+1}$, and the corresponding estimator with minimum risk among all estimators of the form bS^2 is $\tilde{S}^2 = \frac{n-1}{n+1} S^2$.

Example 17 ([25]). We consider the problem of estimating a population variance σ^2 using an estimator of the form $\delta_b = bS^2$, where S^2 is the sample variance and b is a constant.

Let $X_1, \dots, X_n$ be a random sample from a population with positive, finite variance σ^2 . To evaluate the performance of such estimators, we employ the loss function introduced by Stein:

$$L(\sigma^2, a) = \frac{a}{\sigma^2} - 1 - \log \frac{a}{\sigma^2},$$

which has several desirable properties. Unlike squared error loss, this loss penalizes both overestimation and underestimation symmetrically and severely, especially in extreme cases. Specifically, the loss is zero when $a = \sigma^2$, and for any fixed σ^2 , it diverges to infinity as $a \rightarrow 0$ or $a \rightarrow \infty$. This makes it particularly suitable for variance estimation, where underestimation typically incurs less penalty under standard loss functions.

This loss function also arises naturally from the likelihood function for σ^2 in the normal case, providing a coherent decision-theoretic framework that aligns well with likelihood principles. For the estimator $\delta_b = bS^2$, the associated risk function is given by:

$$\begin{aligned} R(\sigma^2, \delta_b) &= \mathbb{E} \left(\frac{bS^2}{\sigma^2} - 1 - \log \frac{bS^2}{\sigma^2} \right) \\ &= b \mathbb{E} \left(\frac{S^2}{\sigma^2} \right) - 1 - \mathbb{E} \log \left(\frac{bS^2}{\sigma^2} \right) \\ &= b - \log b - 1 - \mathbb{E} \log \left(\frac{S^2}{\sigma^2} \right), \end{aligned}$$

where we have used the fact that $\mathbb{E}(S^2/\sigma^2) = 1$.

Note that the term $\mathbb{E} \log(S^2/\sigma^2)$ depends on σ^2 and possibly other population characteristics, but it is independent of b . Hence, minimizing the risk $R(\sigma^2, \delta_b)$ with respect to b reduces to minimizing the function $b - \log b$. The minimum occurs at $b = 1$, implying that the risk is minimized when $\delta_1 = S^2$.

Therefore, among estimators of the form bS^2 , the sample variance S^2 is the one with uniformly minimal risk under Stein's loss.

Depending on the context, the minimization of the risk function $R(a, \theta)$ can be interpreted in different ways. In the *Frequentist* paradigm, for each fixed θ , we choose a to minimize $R(a, \theta)$, and in the *Bayesian* paradigm, if θ is treated as a random variable with prior distribution $\pi(\theta)$, then we aim to minimize the *Bayes risk*:

$$\hat{a} = \arg \min_{a \in \mathcal{A}} \int R(a, \theta) \pi(\theta) d\theta.$$

Definition 12 (Loss, decision rule, Bayes risk). Let $L(a, \theta)$ be a loss function for actions $a \in \mathcal{A}$ and parameter $\theta \in \Theta$. A (measurable) decision rule (estimator) is a mapping $\delta : \mathcal{X} \rightarrow \mathcal{A}$ that assigns an action $\delta(\mathbf{x})$ to each observed data $\mathbf{x} \in \mathcal{X}$. Given a prior density (or mass function) $\pi(\theta)$ and the sampling density $f(\mathbf{x} | \theta)$, the Bayes risk of δ is

$$r(\delta) = \mathbb{E}_\pi[R(\delta(\mathbf{X}), \Theta)] = \int_{\mathcal{X}} \int_{\Theta} L(\delta(\mathbf{x}), \theta) f(\mathbf{x} | \theta) \pi(\theta) d\theta d\mathbf{x},$$

provided the integrals exist.

Proposition 1 (Bayes rule as posterior expected loss minimizer). *Minimizing the Bayes risk*

$$r(\delta) = \int_{\mathcal{X}} \int_{\Theta} L(\delta(\mathbf{x}), \theta) f(\mathbf{x} | \theta) \pi(\theta) d\theta d\mathbf{x}$$

over all decision rules $\delta : \mathcal{X} \rightarrow \mathcal{A}$ is equivalent to, for (almost) every $\mathbf{x}$, choosing

$$\delta^*(\mathbf{x}) \in \arg \min_{a \in \mathcal{A}} \int_{\Theta} L(a, \theta) \pi(\theta | \mathbf{x}) d\theta,$$

that is, minimizing the posterior expected loss pointwise.

Proof. Let

$$m(\mathbf{x}) = \int_{\Theta} f(\mathbf{x} | \theta) \pi(\theta) d\theta$$

denote the marginal (predictive) density of $\mathbf{X}$. By Bayes' theorem, for $m(\mathbf{x}) > 0$ the posterior distribution is

$$\pi(\theta | \mathbf{x}) = \frac{f(\mathbf{x} | \theta) \pi(\theta)}{m(\mathbf{x})}.$$

Starting from the Bayes risk and using Fubini's theorem, we can write:

$$\begin{aligned} r(\delta) &= \int_{\mathcal{X}} \int_{\Theta} L(\delta(\mathbf{x}), \theta) f(\mathbf{x} | \theta) \pi(\theta) d\theta d\mathbf{x} \\ &= \int_{\mathcal{X}} \left[\int_{\Theta} L(\delta(\mathbf{x}), \theta) \pi(\theta | \mathbf{x}) d\theta \right] m(\mathbf{x}) d\mathbf{x}. \end{aligned}$$

Define the posterior expected loss for action a given $\mathbf{x}$:

$$\rho(a | \mathbf{x}) = \int_{\Theta} L(a, \theta) \pi(\theta | \mathbf{x}) d\theta.$$

Then the Bayes risk becomes

$$r(\delta) = \int_{\mathcal{X}} \rho(\delta(\mathbf{x}) | \mathbf{x}) m(\mathbf{x}) d\mathbf{x}.$$

Since $m(\mathbf{x}) \geq 0$, minimizing the integral over all decision rules δ reduces to minimizing the integrand pointwise (for $m(\mathbf{x})$ -almost every $\mathbf{x}$). Therefore, any Bayes decision rule δ^* must satisfy

$$\delta^*(\mathbf{x}) \in \arg \min_{a \in \mathcal{A}} \rho(a | \mathbf{x}) = \arg \min_{a \in \mathcal{A}} \int_{\Theta} L(a, \theta) \pi(\theta | \mathbf{x}) d\theta.$$

If $m(\mathbf{x}) = 0$, the choice of $\delta(\mathbf{x})$ does not affect $r(\delta)$. □

Definition 13 (Posterior expected loss). For observed data $\mathbf{x} \in \mathcal{X}$, the posterior expected loss of choosing action $a \in \mathcal{A}$ given $\mathbf{x}$ is defined as

$$\rho(a | \mathbf{x}) = \mathbb{E}[L(a, \Theta) | \mathbf{X} = \mathbf{x}] = \int_{\Theta} L(a, \theta) \pi(\theta | \mathbf{x}) d\theta.$$

The action a^* that minimizes this posterior expected loss is called the Bayes action.

Example 18. Let $X_1, X_2, \dots, X_n$ be a random sample with joint probability mass or density function $f_{\mathbf{X}}(\mathbf{x} | \theta)$, where the parameter θ belongs to the parameter space $\Theta = \{\theta_1, \theta_2\}$. We aim to estimate the true value of θ using the following loss function:

		True value of θ	
		θ_1	θ_2
Estimated value a	θ_1	L_{11}	L_{12}
	θ_2	L_{21}	L_{22}

Since we typically assume that the losses for a correct estimate are smaller than those for an incorrect estimate, we assume the following conditions hold:

$$L_{11} < L_{21} \quad \text{and} \quad L_{22} < L_{12}.$$

The prior distribution of θ is given by:

$$\pi(\theta_1) = \xi, \quad \pi(\theta_2) = 1 - \xi,$$

where $0 < \xi < 1$. The goal is to find the Bayes estimator of θ with respect to the above loss function.

In this case, let $x_1, x_2, \dots, x_n$ be a realization from the distribution with joint probability mass or density function $f_{\mathbf{X}}(\mathbf{x} | \theta)$, the posterior distribution is as follows. Using Bayes' theorem, the posterior distribution is:

$$\pi(\theta_i | \mathbf{x}) = \frac{f_{\mathbf{X}}(\mathbf{x} | \theta_i) \cdot \pi(\theta_i)}{m(\mathbf{x})},$$

where the marginal likelihood is:

$$m(\mathbf{x}) = f_{\mathbf{X}}(\mathbf{x} | \theta_1) \cdot \xi + f_{\mathbf{X}}(\mathbf{x} | \theta_2) \cdot (1 - \xi).$$

Explicitly:

$$\begin{aligned} \pi(\theta_1 | \mathbf{x}) &= \frac{f_{\mathbf{X}}(\mathbf{x} | \theta_1) \cdot \xi}{f_{\mathbf{X}}(\mathbf{x} | \theta_1) \cdot \xi + f_{\mathbf{X}}(\mathbf{x} | \theta_2) \cdot (1 - \xi)}, \\ \pi(\theta_2 | \mathbf{x}) &= 1 - \pi(\theta_1 | \mathbf{x}) = \frac{f_{\mathbf{X}}(\mathbf{x} | \theta_2) \cdot (1 - \xi)}{f_{\mathbf{X}}(\mathbf{x} | \theta_1) \cdot \xi + f_{\mathbf{X}}(\mathbf{x} | \theta_2) \cdot (1 - \xi)}. \end{aligned}$$

The Bayes estimator minimizes the posterior expected loss:

$$\begin{aligned}\mathbb{E}[L(\theta_1, \theta) \mid x] &= L_{11} \cdot \pi(\theta_1 \mid \mathbf{x}) + L_{12} \cdot \pi(\theta_2 \mid \mathbf{x}), \\ \mathbb{E}[L(\theta_2, \theta) \mid x] &= L_{21} \cdot \pi(\theta_1 \mid \mathbf{x}) + L_{22} \cdot \pi(\theta_2 \mid \mathbf{x}).\end{aligned}$$

The Bayes rule is:

$$\hat{\theta}_{\text{Bayes}}(\mathbf{x}) = \begin{cases} \theta_1 & \text{if } \mathbb{E}[L(\theta_1, \theta) \mid x] < \mathbb{E}[L(\theta_2, \theta) \mid x], \\ \theta_2 & \text{otherwise.} \end{cases}$$

As a special case consider 0-1 Loss, then the Bayes rule becomes:

$$\hat{\theta}_{\text{Bayes}}(\mathbf{x}) = \begin{cases} \theta_1 & \text{if } \pi(\theta_2 \mid \mathbf{x}) > \pi(\theta_1 \mid \mathbf{x}) \\ \theta_2 & \text{otherwise} \end{cases}.$$

Proposition 2 (Bayes estimator under squared error loss). *Let the loss function be the squared error*

$$L(a, \theta) = (a - \theta)^2.$$

Then the Bayes estimator, i.e. the decision rule that minimizes the posterior expected loss, is the posterior mean:

$$\delta^*(\mathbf{x}) = \mathbb{E}[\Theta \mid \mathbf{X} = \mathbf{x}] = \int_{\Theta} \theta \pi(\theta \mid \mathbf{x}) d\theta.$$

Proof. For fixed observed data $\mathbf{x}$, the posterior expected loss is

$$\rho(a \mid \mathbf{x}) = \int_{\Theta} (a - \theta)^2 \pi(\theta \mid \mathbf{x}) d\theta.$$

Expanding the square gives

$$\rho(a \mid \mathbf{x}) = a^2 - 2a \int_{\Theta} \theta \pi(\theta \mid \mathbf{x}) d\theta + \int_{\Theta} \theta^2 \pi(\theta \mid \mathbf{x}) d\theta.$$

The last term does not depend on a . Differentiating with respect to a and setting the derivative to zero yields

$$2a - 2 \int_{\Theta} \theta \pi(\theta \mid \mathbf{x}) d\theta = 0.$$

Thus, the minimizer is

$$a = \int_{\Theta} \theta \pi(\theta \mid \mathbf{x}) d\theta = \mathbb{E}[\Theta \mid \mathbf{X} = \mathbf{x}],$$

which proves that the Bayes estimator under squared error loss is the posterior mean. $\square$

2.8 Exercises

1. **Bayes' theorem basics** A factory produces items of which 5% are defective. A test correctly detects a defective item 90% of the time but also incorrectly classifies a good item as defective 10% of the time.
 - (a) If an item tests positive (defective), what is the posterior probability it is actually defective?
 - (b) Interpret the result in the Bayesian framework.
2. **Beta–Binomial conjugacy** Suppose you observe $y = 7$ successes out of $n = 10$ Bernoulli trials.
 - (a) Assume a Beta(1, 1) prior. Derive the posterior distribution of the success probability θ .
 - (b) Compute the posterior mean, mode (MAP), and a 95% credible interval.
 - (c) Repeat with a more informative prior, Beta(5, 5), and compare results.
3. **Posterior with Normal likelihood and prior** Let $y_1, \dots, y_n \sim \text{Normal}(\mu, \sigma^2)$, with known σ^2 . Assume a prior $\mu \sim \text{Normal}(\mu_0, \tau^2)$.
 - (a) Derive the posterior distribution of μ .
 - (b) Show that it is also Normal and identify its mean and variance.
 - (c) Compute the posterior for $n = 20, \bar{y} = 2.3, \sigma^2 = 1, \mu_0 = 0, \tau^2 = 4$.
4. **Credible interval vs. confidence interval** Simulate $n = 100$ Bernoulli trials with true probability $\theta = 0.6$.
 - (a) Fit a Bayesian model with Beta(1, 1) prior and compute a 95% credible interval.
 - (b) Compute the corresponding frequentist 95% confidence interval.
 - (c) Compare and interpret the difference in interpretation.
5. **Sufficiency in Bayesian inference** Show that for Bernoulli trials $y_1, \dots, y_n \sim \text{Bernoulli}(\theta)$, the sufficient statistic for θ is $\sum_{i=1}^n y_i$.
 - (a) Write down the posterior distribution for θ using only the sufficient statistic.
 - (b) Explain why sufficiency simplifies Bayesian computation.
6. **Decision theory and Bayesian estimators** Suppose the posterior distribution of a parameter θ is Normal(5, 1).
 - (a) Under squared-error loss, what is the Bayes estimator?

- (b) Under absolute-error loss, what is the Bayes estimator?
 - (c) Discuss how the choice of loss function influences decision-making in practice.
7. **Prior sensitivity** Choose one of the earlier models (e.g., binomial with beta prior or normal with normal prior).
- (a) Fit the model under three different priors: weakly informative, informative, and strongly informative.
 - (b) Compare the posterior distributions.
 - (c) Discuss how the prior affects inference, especially with small sample sizes.
8. **Improper prior and posterior propriety** Consider $y_1, \dots, y_n \sim \text{Poisson}(\lambda)$.
- (a) Assume an improper prior $\pi(\lambda) \propto 1/\lambda$ for $\lambda > 0$.
 - (b) Derive the posterior distribution of λ .
 - (c) For which values of n is the posterior proper (i.e., integrates to 1)?
9. **Hierarchical prior** Suppose exam scores follow $y_i \sim \text{Normal}(\mu, \sigma^2)$ with known $\sigma^2 = 1$. Assume a hierarchical prior:

$$\mu \mid \tau^2 \sim \text{Normal}(0, \tau^2), \quad \tau^2 \sim \text{Inverse-Gamma}(2, 2).$$

- (a) Write down the joint posterior of (μ, τ^2) .
 - (b) Explain how hierarchical priors allow more flexible modeling of uncertainty.
 - (c) Discuss why analytical solutions are not straightforward and what computational approaches can be used.
10. **Conjugate prior for the Poisson model** Let $X_1, X_2, \dots, X_n$ be independent observations from a Poisson distribution with unknown rate $\theta > 0$:

$$X_i \sim \text{Poisson}(\theta), \quad i = 1, \dots, n.$$

Assume a Gamma prior for θ :

$$\theta \sim \text{Gamma}(\alpha, \beta),$$

where $\alpha > 0$ is the shape parameter and $\beta > 0$ is the rate parameter.

- (a) Generate a random sample of size $n = 50$ from a Poisson distribution with $\theta = 4$.
- (b) Set the prior hyperparameters to $\alpha = 2$ and $\beta = 1$.
- (c) Derive the posterior distribution parameters α_n and β_n for θ .
- (d) Draw a histogram of the simulated Poisson data.

- (e) Draw a histogram of 1000 samples from the posterior distribution of θ .
- (f) Overlay the posterior density on the histogram and indicate the posterior mean.
- (g) Compute a 95% credible interval for θ based on the posterior distribution.

11. **Bayes estimator under absolute error loss** Show that under the absolute error loss function

$$L(a, \theta) = |a - \theta|,$$

the Bayes estimator is posterior median of θ given $\mathbf{X} = \mathbf{x}$.